

# An Introduction to First-Principles Electron Dynamics Calculations Based on Time-Dependent Density Functional Theory

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## Preface

These notes were prepared for students conducting research in the Sato Group of the Condensed Matter Theory Laboratory, Graduate School of Science, Tohoku University, with the aim of introducing the fundamentals of first-principles electron dynamics calculations based on time-dependent density functional theory. They are also intended to provide a first exposure to practical density functional theory and time-dependent density functional theory calculations for simple molecules and solids using first-principles simulation software, thereby serving as preparation for theoretical studies of light-matter interactions. These notes are still in draft form, and their contents are updated from time to time. For the latest version, please consult the note at the URL below.

[https://shunsuke-sato.github.io/page/etc/lecture\\_notes/LectureNoteForTDDFT\\_en.pdf](https://shunsuke-sato.github.io/page/etc/lecture_notes/LectureNoteForTDDFT_en.pdf)

# 1 Introduction: Time-Dependent Schrödinger Equation for Many-Electron Systems

In these notes, we consider first-principles electron dynamics calculations based on time-dependent density functional theory (TDDFT). The starting point is the many-body time-dependent Schrödinger equation that governs the time evolution of a many-electron system. In this section, we first review the form of this exact many-body problem and clarify which quantities must be specified in order to determine the time evolution. This serves as preparation for introducing TDDFT in the next section, where the density is taken as the basic variable.

The main object considered in these notes is a quantum many-body system described by the time-dependent Schrödinger equation for an  $N$ -particle system of the form

$$i\hbar \frac{\partial}{\partial t} \Psi(\mathbf{r}_1, \sigma_1, \dots, \mathbf{r}_N, \sigma_N, t) = \hat{H}(t) \Psi(\mathbf{r}_1, \sigma_1, \dots, \mathbf{r}_N, \sigma_N, t), \quad (1)$$

$$\hat{H}(t) = \sum_{j=1}^N \left[ \frac{\mathbf{p}_j^2}{2m} + v(\mathbf{r}_j, t) \right] + \frac{1}{2} \sum_{j \neq k} w(|\mathbf{r}_j - \mathbf{r}_k|). \quad (2)$$

Here,  $\mathbf{r}_j$  denotes the position of the  $j$ th particle, and  $\sigma_j$  its spin coordinate. In addition,  $v(\mathbf{r}, t)$  is the external one-body potential, and  $w(r)$  is the two-body interaction potential.

For the many-electron systems we have in mind, these potentials may, for example, be taken as

$$v(\mathbf{r}, t) = -\frac{e^2}{4\pi\epsilon_0} \sum_a \frac{Z_a}{|\mathbf{r} - \mathbf{R}_a(t)|} - e\phi_{\text{ext}}(\mathbf{r}, t), \quad (3)$$

$$w(r) = \frac{e^2}{4\pi\epsilon_0} \frac{1}{r}. \quad (4)$$

Here,  $Z_a$  and  $\mathbf{R}_a(t)$  denote the nuclear charge and position of the  $a$ th nucleus, respectively, and  $\phi_{\text{ext}}(\mathbf{r}, t)$  is a time-dependent external electromagnetic scalar potential. With the one-body and two-body potentials specified in this way, Eq. (1) describes the dynamics of an  $N$ -electron system moving under the Coulomb potential generated by the nuclei and the external electromagnetic scalar potential.

In principle, if Eq. (1) could be solved directly, one would obtain a detailed description of the dynamics of an electron system driven by an external potential. Through such an analysis, one could also describe theoretically a wide variety of physical phenomena arising from light-matter interactions. In practice, however, solving the many-body Schrödinger equation directly is generally extremely difficult. The reason is that, as the number of particles increases, the dimension of the space in which the wavefunction is defined grows exponentially.

We therefore regard Eq. (1) as an initial-value problem. Let the wavefunction at the initial time  $t = t_0$  be

$$\Psi(\mathbf{r}_1, \sigma_1, \dots, \mathbf{r}_N, \sigma_N, t_0) = \Psi_0(\mathbf{r}_1, \sigma_1, \dots, \mathbf{r}_N, \sigma_N) \quad (5)$$

Further, throughout these notes we assume the following two points.

- The two-body interaction  $w(r)$  is given in advance and cannot be changed arbitrarily. For example, in an electronic system,  $w(r)$  is the Coulomb interaction itself, and it is not something that can be modified.
- The external one-body potential  $v(\mathbf{r}, t)$  is specified separately for each problem. For example, when one wishes to control an electron system by irradiating it with electromagnetic waves,  $v(\mathbf{r}, t)$  must be chosen appropriately according to the physical problem under consideration.

From this viewpoint, since the two-body interaction  $w(r)$  is fixed, the quantities that must be specified for each problem are the initial wavefunction  $\Psi_0(\mathbf{r}_1, \sigma_1, \dots, \mathbf{r}_N, \sigma_N)$  and the one-body

potential  $v(\mathbf{r}, t)$ . Once these are given, the physical problem is completely defined, and solving Eq. (1) uniquely determines the time-dependent many-body wavefunction  $\Psi(\mathbf{r}_1, \sigma_1, \dots, \mathbf{r}_N, \sigma_N, t)$ . Writing this as

$$\Psi(\mathbf{r}_1, \sigma_1, \dots, \mathbf{r}_N, \sigma_N, t) = \Psi[\Psi_0(\mathbf{r}_1, \sigma_1, \dots, \mathbf{r}_N, \sigma_N), v(\mathbf{r}, t)](\mathbf{r}_1, \sigma_1, \dots, \mathbf{r}_N, \sigma_N, t)$$

shows that the many-body wavefunction is a functional of the initial wavefunction  $\Psi_0(\mathbf{r}_1, \sigma_1, \dots, \mathbf{r}_N, \sigma_N)$  and the one-body potential  $v(\mathbf{r}, t)$ . Moreover, if the wavefunction is a functional of these quantities, then any physical observable computed from it can likewise be expressed as a functional of the same quantities. This viewpoint leads directly to the basic idea of time-dependent density functional theory discussed in the next section. Note also that, in the present problem, once the one-body potential at the initial time ( $t = t_0$ ) is specified, the many-body Hamiltonian [Eq. (2)] is fixed. Its ground state is then a functional of the one-body potential at the initial time. In many problems, the initial wavefunction is taken to be this ground state. Therefore, if one restricts the initial condition to the ground state, the many-body wavefunction  $\Psi(\mathbf{r}_1, \sigma_1, \dots, \mathbf{r}_N, \sigma_N, t)$  may be regarded as a functional of the one-body potential  $v(\mathbf{r}, t)$  alone, and all physical observables likewise become functionals of  $v(\mathbf{r}, t)$ .

## 2 Fundamentals of Time-Dependent Density Functional Theory

In the previous section, we confirmed that once the initial wavefunction and the one-body potential are given, the time evolution of the many-body system is uniquely determined. In this section, we introduce the basic idea of time-dependent density functional theory (TDDFT), which replaces the description in terms of the wavefunction by one in terms of the particle density. We first explain, through the Runge–Gross theorem, why the density can be adopted as the basic variable, and then introduce the Kohn–Sham mapping that connects this idea to practical calculations.

### 2.1 Runge–Gross theorem

In density functional theory (DFT) for ground states, the Hohenberg–Kohn theorem, which establishes a one-to-one correspondence between the one-body potential and the ground-state density, plays a central role. The counterpart for time-dependent problems is the Runge–Gross theorem. It states that, for a fixed initial state, there is a one-to-one correspondence between the one-body potential  $v(\mathbf{r}, t)$  and the particle density  $\rho(\mathbf{r}, t)$ . Therefore, the density can be used as a basic variable in place of the wavefunction.

In what follows, we assume that the many-body wavefunction  $\Psi(\mathbf{r}_1, \sigma_1, \dots, \mathbf{r}_N, \sigma_N, t)$  is normalized, and define the particle density  $\rho(\mathbf{r}, t)$  and the current density  $\mathbf{j}(\mathbf{r}, t)$  by

$$\rho(\mathbf{r}, t) = N \sum_{\sigma_1, \dots, \sigma_N} \int d\mathbf{r}_2 \cdots d\mathbf{r}_N |\Psi(\mathbf{r}, \sigma_1, \dots, \mathbf{r}_N, \sigma_N, t)|^2, \quad (6)$$

$$\mathbf{j}(\mathbf{r}, t) = N \sum_{\sigma_1, \dots, \sigma_N} \text{Re} \left[ \int d\mathbf{r}_2 \cdots d\mathbf{r}_N \Psi^*(\mathbf{r}, \sigma_1, \dots, \mathbf{r}_N, \sigma_N, t) \frac{\mathbf{p}}{m} \Psi(\mathbf{r}, \sigma_1, \dots, \mathbf{r}_N, \sigma_N, t) \right] \quad (7)$$

Here,  $\mathbf{p} = -i\hbar\nabla$  is the momentum operator acting on the coordinate  $\mathbf{r}$ . These quantities satisfy the continuity equation expressing particle-number conservation:

$$\frac{\partial}{\partial t} \rho(\mathbf{r}, t) + \nabla \cdot \mathbf{j}(\mathbf{r}, t) = 0 \quad (8)$$

As in the previous section, we assume that the two-body interaction  $w(r)$  is fixed and that the many-body wavefunction evolves in time according to Eq. (1). Let the initial condition at time  $t = t_0$  be written as

$$\Psi(\mathbf{r}_1, \sigma_1, \dots, \mathbf{r}_N, \sigma_N, t_0) = \Psi_0(\mathbf{r}_1, \sigma_1, \dots, \mathbf{r}_N, \sigma_N) \quad (9)$$

Since the initial wavefunction  $\Psi_0$  and the two-body interaction  $w(r)$  are fixed, the only quantity that can be chosen freely from one problem to another is the one-body potential  $v(\mathbf{r}, t)$ .

The content of the Runge–Gross theorem is that, under these conditions, a one-to-one correspondence holds between the one-body potential  $v(\mathbf{r}, t)$  and the density  $\rho(\mathbf{r}, t)$ .<sup>1</sup> Below, assuming that the potential is Taylor expandable in the neighborhood of  $t = t_0$ , we follow the outline of the proof.

First, consider two potentials  $v(\mathbf{r}, t)$  and  $v'(\mathbf{r}, t)$  that do not coincide up to a function of time alone. That is,

$$v(\mathbf{r}, t) - v'(\mathbf{r}, t) \neq C(t) \quad (10)$$

Then there exists a smallest nonnegative integer  $k$  satisfying

$$\left. \frac{\partial^k}{\partial t^k} [v(\mathbf{r}, t) - v'(\mathbf{r}, t)] \right|_{t=t_0} \neq \text{const.} \quad (11)$$

---

<sup>1</sup>Here, potentials that differ only by a function of time are regarded as identical.

Since the two systems start from the same initial wavefunction  $\Psi_0$  and share the same two-body interaction, the time derivative of the difference between their current densities satisfies

$$\frac{\partial}{\partial t} [\mathbf{j}(\mathbf{r}, t) - \mathbf{j}'(\mathbf{r}, t)] = -\frac{1}{m} [\rho(\mathbf{r}, t) \nabla v(\mathbf{r}, t) - \rho'(\mathbf{r}, t) \nabla v'(\mathbf{r}, t)] \quad (12)$$

Furthermore, differentiating both sides  $k$  times with respect to  $t$  and using the fact that the two wavefunctions coincide with the initial condition in Eq. (9) at  $t = t_0$ , we obtain

$$\frac{\partial^{k+1}}{\partial t^{k+1}} [\mathbf{j}(\mathbf{r}, t) - \mathbf{j}'(\mathbf{r}, t)] \Big|_{t=t_0} = -\frac{\rho(\mathbf{r}, t_0)}{m} \nabla \frac{\partial^k}{\partial t^k} [v(\mathbf{r}, t) - v'(\mathbf{r}, t)] \Big|_{t=t_0} \neq 0 \quad (13)$$

Therefore, if the two potentials are different, the corresponding current densities must also differ for  $t > t_0$ .

Next, differentiate the density difference  $\rho(\mathbf{r}, t) - \rho'(\mathbf{r}, t)$   $(k+2)$  times with respect to  $t$ . Using Eq. (8) together with Eq. (13), one finds

$$\frac{\partial^{k+2}}{\partial t^{k+2}} [\rho(\mathbf{r}, t) - \rho'(\mathbf{r}, t)] \Big|_{t=t_0} = \frac{1}{m} \nabla \cdot \left( \rho(\mathbf{r}, t_0) \nabla \frac{\partial^k}{\partial t^k} [v(\mathbf{r}, t) - v'(\mathbf{r}, t)] \Big|_{t=t_0} \right) \quad (14)$$

Thus, if one can show that the right-hand side of Eq. (14) is nonzero, it follows that two different potentials give rise to two different densities.

To this end, suppose that for a function  $u(\mathbf{r}) \neq \text{const.}$ ,

$$\nabla \cdot (\rho(\mathbf{r}, t_0) \nabla u(\mathbf{r})) = 0 \quad (15)$$

holds. Multiplying by  $u(\mathbf{r})$ , integrating over space, and performing integration by parts, we obtain

$$\begin{aligned} 0 &= \int d\mathbf{r} u(\mathbf{r}) \nabla \cdot (\rho(\mathbf{r}, t_0) \nabla u(\mathbf{r})) \\ &= - \int d\mathbf{r} \rho(\mathbf{r}, t_0) |\nabla u(\mathbf{r})|^2 + \frac{1}{2} \int_S d\mathbf{S} \cdot \rho(\mathbf{r}, t_0) \nabla u^2(\mathbf{r}) \end{aligned} \quad (16)$$

If the initial density  $\rho(\mathbf{r}, t_0)$  decays sufficiently rapidly at large distances, the surface integral vanishes. One then obtains  $\rho(\mathbf{r}, t_0) |\nabla u(\mathbf{r})|^2 = 0$ , which contradicts  $u(\mathbf{r}) \neq \text{const.}$ . Therefore,

$$\nabla \cdot (\rho(\mathbf{r}, t_0) \nabla u(\mathbf{r})) \neq 0 \quad (17)$$

and hence the right-hand side of Eq. (14) is nonzero. It follows that densities evolving under different potentials must necessarily be different.

We have thus shown that, when time evolution is generated from a given initial wavefunction, there is a one-to-one correspondence between the one-body potential  $v(\mathbf{r}, t)$  and the density  $\rho(\mathbf{r}, t)$ . This is the Runge–Gross theorem, which provides the starting point of TDDFT. Combined with the discussion in the previous section, it implies that the many-body wavefunction

$$\Psi[\Psi_0(\mathbf{r}_1, \sigma_1, \dots, \mathbf{r}_N, \sigma_N), v(\mathbf{r}, t)](\mathbf{r}_1, \sigma_1, \dots, \mathbf{r}_N, \sigma_N, t)$$

can also be expressed as a functional of the initial wavefunction  $\Psi_0$  and the density  $\rho(\mathbf{r}, t)$ :

$$\Psi[\Psi_0(\mathbf{r}_1, \sigma_1, \dots, \mathbf{r}_N, \sigma_N), \rho(\mathbf{r}, t)](\mathbf{r}_1, \sigma_1, \dots, \mathbf{r}_N, \sigma_N, t)$$

Therefore, any physical quantity can be described as a functional of the initial wavefunction  $\Psi_0$  and the density  $\rho(\mathbf{r}, t)$ .

Moreover, in many applications the initial wavefunction is taken to be the ground state of the initial Hamiltonian. In that case, the Hohenberg–Kohn theorem implies a one-to-one correspondence between the one-body potential  $v(\mathbf{r}, t_0)$  at the initial time  $t = t_0$  and the ground-state density  $\rho_{\text{gs}}(\mathbf{r})$ . Hence, the initial wavefunction itself may be regarded as a functional of the initial density, and the many-body wavefunction can then be treated as a functional of the density  $\rho(\mathbf{r}, t)$  alone. This is the reason why the density is adopted as the basic variable in TDDFT.

## 2.2 Kohn–Sham Mapping and the Time-Dependent Kohn–Sham Equation

The Runge–Gross theorem tells us that a many-body system can be described using the density as its basic variable. By itself, however, this theorem does not yet tell us how to compute the time evolution of an interacting many-body system in practice. What makes TDDFT a practical computational method is the Kohn–Sham mapping, which maps an interacting system onto a noninteracting one while preserving the density.

Let us call the original many-body system of interest the **interacting system**. It is specified by the initial state  $|\Psi_0\rangle$ , the one-body potential  $v(\mathbf{r}, t)$ , and the two-body interaction  $w(\mathbf{r})$ , and its density is denoted by  $\rho(\mathbf{r}, t)$ . In contrast, we introduce a noninteracting system with the same number of particles and call it the **Kohn–Sham system**. In the Kohn–Sham system, each particle moves under an effective one-body potential  $v_{\text{KS}}(\mathbf{r}, t)$ , and its density is denoted by  $\rho_{\text{KS}}(\mathbf{r}, t)$ .

The basic idea of TDDFT is to construct a Kohn–Sham system that yields the same density as the interacting system. That is, we consider a noninteracting system such that  $\rho_{\text{KS}}(\mathbf{r}, t) = \rho(\mathbf{r}, t)$ . Strictly speaking, the initial state  $|\Phi_0\rangle$  of the Kohn–Sham system must be chosen so that not only the initial density  $\rho(\mathbf{r}, t_0)$  but also its time derivative (or equivalently the initial current density) is consistent with that of the interacting system. Once this condition is satisfied, the density of the interacting system can be reproduced from single-particle orbitals of the noninteracting system.

Since the Runge–Gross theorem also holds for the noninteracting system, fixing the initial state  $|\Phi_0\rangle$  establishes a one-to-one correspondence between the density  $\rho_{\text{KS}}(\mathbf{r}, t)$  and the Kohn–Sham potential  $v_{\text{KS}}(\mathbf{r}, t)$ . Therefore, if one can find  $v_{\text{KS}}(\mathbf{r}, t)$  that yields the same density as the interacting system, the interacting many-body problem can be replaced by a noninteracting problem under an effective one-body potential. This correspondence is called the Kohn–Sham mapping.

In many applications, the system at the initial time  $t_0$  is prepared in the ground state of the one-body potential  $v(\mathbf{r}, t_0)$ . In that case, it is natural to choose the initial states of the interacting system and the Kohn–Sham system as the ground states of their respective initial Hamiltonians. Furthermore, by the Hohenberg–Kohn theorem, once the initial ground-state density is given, the corresponding initial state is also determined. Therefore, if one restricts attention to time evolution starting from a ground state, the initial-state dependence appearing in the Kohn–Sham mapping can effectively be absorbed into the initial density.

If the initial state of the Kohn–Sham system is represented by a single Slater determinant, then the time-evolved state is also represented by a single Slater determinant. In this case, the single-particle orbitals  $\phi_k(\mathbf{r}, \sigma, t)$  forming the determinant are called Kohn–Sham orbitals, and the density is written as  $\rho(\mathbf{r}, t) = \sum_k \sum_\sigma |\phi_k(\mathbf{r}, \sigma, t)|^2$ . Each orbital satisfies the following time-dependent Kohn–Sham equation:

$$i\hbar \frac{\partial}{\partial t} \phi_k(\mathbf{r}, \sigma, t) = \left[ \frac{\mathbf{p}^2}{2m} + v_{\text{KS}}(\mathbf{r}, t) \right] \phi_k(\mathbf{r}, \sigma, t). \quad (18)$$

Thus, instead of solving the many-body time-dependent Schrödinger equation (1) directly, the practical task in TDDFT is to solve the single-particle equation given by Eq. (18).

In actual calculations, the Kohn–Sham potential is decomposed as

$$v_{\text{KS}}(\mathbf{r}, t) = v(\mathbf{r}, t) + v_{\text{H}}(\mathbf{r}, t) + v_{\text{XC}}(\mathbf{r}, t) \quad (19)$$

Here,  $v(\mathbf{r}, t)$  is the external one-body potential,  $v_{\text{H}}(\mathbf{r}, t)$  is the Hartree potential representing the classical mean-field interaction, and  $v_{\text{XC}}(\mathbf{r}, t)$  is the exchange-correlation potential that accounts for the remaining many-body effects not described by the Hartree term. The Hartree potential is defined by

$$v_{\text{H}}(\mathbf{r}, t) = \int d\mathbf{r}' w(|\mathbf{r} - \mathbf{r}'|) \rho(\mathbf{r}', t) \quad (20)$$



On the other hand, the exchange-correlation potential is defined as

$$v_{\text{XC}}(\mathbf{r}, t) = v_{\text{KS}}(\mathbf{r}, t) - v(\mathbf{r}, t) - v_{\text{H}}(\mathbf{r}, t) \quad (21)$$

and is chosen so that the density of the Kohn–Sham system coincides with that of the interacting system.

Up to this point, the discussion has been exact. However, explicit expressions for the exact  $v_{\text{KS}}(\mathbf{r}, t)$  and  $v_{\text{XC}}(\mathbf{r}, t)$  are generally not known, so in practical TDDFT calculations one must approximate  $v_{\text{XC}}(\mathbf{r}, t)$  in some way. In the latter part of these notes, we use one of the most basic approximations, the adiabatic local density approximation (Adiabatic Local Density Approximation; ALDA). In this approximation, the exchange-correlation potential at each time is approximated as a function of the instantaneous electron density.

In this way, TDDFT allows the dynamics of an interacting many-body system to be treated as the time evolution of a noninteracting Kohn–Sham system that reproduces the same density. Accordingly, in what follows, the central task is to choose an appropriate approximation for the exchange-correlation potential and solve Eq. (18) numerically. In the next section, we introduce the pseudopotential method, which plays an important role in its implementation.

### 3 Pseudopotential Method

In the previous section, we saw that the central task in TDDFT calculations is to solve the time-dependent Kohn–Sham equation self-consistently. If one attempts to implement this directly, however, the steep behavior of the electronic wavefunctions near the nuclei gives rise to serious numerical difficulties. In this section, we introduce the **pseudopotential method**, which alleviates this difficulty while retaining, as much as possible, the essential physics of the valence electrons.

To describe an electron system driven by an external electromagnetic potential, it is natural to employ a one-body potential such as that given in Eq. (3). Calculations that explicitly treat even the electrons near the nuclei in this way are called **all-electron calculations**. However, the Coulomb potential generated by a nucleus diverges at the nuclear position, and as a result the wavefunction exhibits a sharp structure in its vicinity. For this reason, all-electron calculations are not easy to handle numerically.<sup>2</sup>

On the other hand, in many molecular and solid-state problems, the main role is played by the valence electrons, which are weakly bound to the nuclei, whereas the core electrons, which are deeply bound, can often be regarded as remaining almost unchanged from the ground state of the isolated atom without significant loss of accuracy. From this viewpoint, the pseudopotential method greatly reduces the computational cost by

- freezing the degrees of freedom of the core electrons, and
- replacing the valence-electron wavefunctions by functions that are smoother near the nuclei.

This is the basic idea of the pseudopotential method. In what follows, we explain the idea using the **norm-conserving pseudopotential** of Troullier–Martins [1] as an example among pseudopotentials constructed starting from atomic DFT calculations. For further details, see, for example, Ref. [2].

First, consider the radial Kohn–Sham equation for a spherically symmetric atom:

$$\left[ -\frac{\hbar^2}{2m} \frac{d^2}{dr^2} + \frac{\hbar^2 l(l+1)}{2m} + v_{\text{KS}}^{\text{AE}}[\rho^{\text{AE}}(r)](r) \right] \chi_{nl}^{\text{AE}}(r) = \epsilon_{nl} \chi_{nl}^{\text{AE}}(r). \quad (22)$$

Here,  $n$  denotes the principal quantum number and  $l$  the orbital angular momentum quantum number. In addition,  $\rho^{\text{AE}}(r)$  is the density obtained in an all-electron calculation, and the Kohn–Sham potential  $v_{\text{KS}}^{\text{AE}}[\rho^{\text{AE}}(r)](r)$  can be written as the sum of the nuclear Coulomb potential, the Hartree potential  $v_{\text{H}}[\rho^{\text{AE}}(r)](r)$ , and the exchange-correlation potential  $v_{\text{xc}}[\rho^{\text{AE}}(r)](r)$ :

$$v_{\text{KS}}^{\text{AE}}[\rho^{\text{AE}}(r)](r) = -\frac{eZ}{4\pi\epsilon_0} \frac{1}{r} + v_{\text{H}}[\rho^{\text{AE}}(r)](r) + v_{\text{xc}}[\rho^{\text{AE}}(r)](r) \quad (23)$$

Furthermore, by combining the radial wavefunction  $\chi_{nl}(r)$  with the spherical harmonic  $Y_{lm}(\theta, \phi)$ , one obtains a Kohn–Sham orbital

$$\phi_{nlm}(\mathbf{r}) = \frac{\chi_{nl}(r)}{r} Y_{lm}(\theta, \phi) \quad (24)$$

The first step in constructing a pseudopotential is to divide the atomic Kohn–Sham orbitals into **core electrons** and **valence electrons**. Core electrons are deeply bound and spatially localized near the nucleus, so they are assumed not to differ greatly from those of the isolated atom even in molecules and solids. By contrast, valence electrons are weakly bound and spatially extended, and they participate directly in bonding, excitation, and response to external fields; they are therefore the degrees of freedom that should be treated explicitly in actual calculations. Where one draws the boundary between core and valence is an important choice in pseudopotential

<sup>2</sup>For example, if the wavefunction is expanded in plane waves, then in order to reproduce such a sharp structure one must include plane waves with very large wave numbers in the expansion, which in turn requires a very large number of basis functions.

design and must be determined appropriately according to the system and physical quantities of interest.

Even after freezing the core degrees of freedom, the all-electron wavefunction of the valence electrons remains steep near the nucleus. We therefore replace it there by a smoother function, while requiring it to coincide with the all-electron wavefunction outside a cutoff radius. Specifically, the pseudowavefunction  $\chi_{nl}^{\text{PP}}(r)$  is defined by

$$\chi_{nl}^{\text{PP}}(r) = \begin{cases} \chi_{nl}^{\text{AE}}(r) & \text{if } r > r_{c,l} \\ r^{l+1} \exp[p(r)] & \text{if } r \leq r_{c,l} \end{cases} \quad (25)$$

Here,  $r_{c,l}$  is the cutoff radius, and outside this radius the pseudowavefunction coincides with the all-electron wavefunction. The function  $p(r)$  is a polynomial, and in the Troullier–Martins method one uses the form

$$p(r) = c_0 + c_2 r^2 + c_4 r^4 + c_6 r^6 + c_8 r^8 + c_{10} r^{10} + c_{12} r^{12} \quad (26)$$

The polynomial coefficients  $c_j$  are determined so as to satisfy the following seven conditions:

- Condition 1. Norm-conservation condition  
 $\int_0^{r_{c,l}} dr |\chi_{nl}^{\text{AE}}(r)|^2 = \int_0^{r_{c,l}} dr |\chi_{nl}^{\text{PP}}(r)|^2$
- Conditions 2.–6.  
 Continuity of the wavefunction and its derivatives up to fourth order at the cutoff radius  $r = r_{c,l}$
- Condition 7.  
 The curvature of the screened pseudopotential  $v_{\text{scr},l}(r)$  at the origin ( $r = 0$ ) is zero:  $v_{\text{scr},l}''(r = 0) = 0$ . Here,  $v_{\text{scr},l}(r)$  will be defined below.

In particular, a pseudopotential constructed so as to satisfy the norm-conservation condition in Condition 1 is called a **norm-conserving pseudopotential**. This means that the pseudowavefunction is designed not only to be smooth but also to preserve, as much as possible, important information contained in the all-electron wavefunction.

Next, define the electron density generated by the valence electrons represented by the pseudowavefunctions  $\chi_{nl}^{\text{PP}}(r)$  as

$$\rho^{\text{PP}}(r) = \sum_{n,l} f_{n,l} |\chi_{nl}^{\text{PP}}(r)|^2 \quad (27)$$

Here,  $f_{n,l}$  is the occupation number of the state. Using this, we define the screened pseudopotential  $v_{\text{scr},l}(r)$  by

$$v_{\text{scr},l}(r) = \begin{cases} v_{\text{KS}}^{\text{AE}}[\rho^{\text{AE}}(r)](r) & \text{if } r > r_{c,l} \\ \epsilon_{nl} + \frac{\hbar^2}{m} \frac{l+1}{r} \frac{p'(r)}{2} + \frac{\hbar^2}{m} \frac{p''(r) + [p'(r)]^2}{2} & \text{if } r \leq r_{c,l} \end{cases} \quad (28)$$

Then the pseudowavefunction  $\chi_{nl}^{\text{PP}}(r)$  satisfies

$$\left[ -\frac{\hbar^2}{2m} \frac{d^2}{dr^2} + \frac{\hbar^2 l(l+1)}{2m} + v_{\text{scr},l}(r) \right] \chi_{nl}^{\text{PP}}(r) = \epsilon_{nl} \chi_{nl}^{\text{PP}}(r) \quad (29)$$

Therefore, once  $v_{\text{scr},l}(r)$  is obtained as the self-consistent field in a DFT calculation for the valence electrons, the resulting orbital functions are indeed pseudowavefunctions.

However,  $v_{\text{scr},l}(r)$  also contains the Hartree potential  $v_{\text{H}}[\rho^{\text{PP}}(r)](r)$  and the exchange-correlation potential  $v_{\text{xc}}[\rho^{\text{PP}}(r)](r)$  generated by the valence electrons. We therefore subtract them and define the pseudopotential exerted by the ion on the valence electrons as

$$v_l^{\text{PP}}(r) = v_{\text{scr},l}(r) - v_{\text{H}}[\rho^{\text{PP}}(r)](r) - v_{\text{xc}}[\rho^{\text{PP}}(r)](r) \quad (30)$$

Since this pseudopotential differs for each angular momentum  $l$ , it must be combined into a single operator before it can be used as an actual one-body potential. To this end, we choose one local potential  $v_{\text{ion,local}}^{\text{PP}}(r)$  and represent the remainder as nonlocal terms:

$$\hat{v}_{\text{ion}}^{\text{PP}} = v_{\text{ion,local}}^{\text{PP}}(r) + \sum_{lm} \Delta v_l^{\text{PP}}(r) \hat{P}_{lm} \quad (31)$$

Here,  $\Delta v_l^{\text{PP}}(r)$  appearing in the nonlocal part is defined by

$$\Delta v_l^{\text{PP}}(r) = v_l^{\text{PP}}(r) - v_{\text{ion,local}}^{\text{PP}}(r) \quad (32)$$

In addition,  $\hat{P}_{lm}$  is the angular-momentum projection operator, defined for an arbitrary function  $f(\mathbf{r})$  by

$$\hat{P}_{lm} f(\mathbf{r}) = Y_{lm}(\theta, \phi) \int d\Omega Y_{lm}^*(\theta, \phi) f(\mathbf{r}) \quad (33)$$

If the pseudopotential in Eq. (31) is used as the one-body potential in the Kohn–Sham equation, then, by construction, the pseudowavefunction  $\chi_{nl}^{\text{PP}}(r)$  is a self-consistent solution. The local potential  $v_{\text{ion,local}}^{\text{PP}}(r)$  may in principle be chosen freely, but in many cases one adopts the pseudopotential  $v_l^{\text{PP}}(r)$  corresponding to a particular angular momentum  $l$ . If the angular momentum used as the reference for the local potential is denoted by  $l_{\text{ref}}$ , then the nonlocal potential  $\Delta v_l^{\text{PP}}(r)$  vanishes for  $r > \max(r_{c,l}, r_{c,l_{\text{ref}}})$  and thus acts only within a finite spatial range.

Equation (31) realizes both the freezing of the core degrees of freedom and the introduction of smooth pseudowavefunctions, but in actual numerical calculations it is computationally disadvantageous to treat the angular-momentum projection operators directly. Therefore, in many practical calculations one uses the Kleinman–Bylander separable form and writes the pseudopotential as

$$\hat{v}_{\text{ion}}^{\text{PP,KB}} = v_{\text{ion,local}}^{\text{PP}}(r) + \sum_{lm} \frac{|\Delta v_l^{\text{PP}}(r) \phi_{lm}^{\text{PP}}\rangle \langle \phi_{lm}^{\text{PP}} \Delta v_l^{\text{PP}}(r)|}{\langle \phi_{lm}^{\text{PP}} | \Delta v_l^{\text{PP}}(r) | \phi_{lm}^{\text{PP}} \rangle} \quad (34)$$

Here, the state vectors  $|\phi_{lm}^{\text{PP}}\rangle$  and  $|\Delta v_l^{\text{PP}}(r) \phi_{lm}^{\text{PP}}\rangle$  are introduced by

$$\langle \mathbf{r} | \phi_{lm}^{\text{PP}} \rangle = \frac{\chi_{nl}^{\text{PP}}(r)}{r} Y_{lm}(\theta, \phi), \quad (35)$$

$$\langle \mathbf{r} | \Delta v_l^{\text{PP}}(r) \phi_{lm}^{\text{PP}} \rangle = \Delta v_l^{\text{PP}}(r) \frac{\chi_{nl}^{\text{PP}}(r)}{r} Y_{lm}(\theta, \phi) \quad (36)$$

In the form of Eq. (34), the nonlocal term can be evaluated more efficiently.

In this way, the pseudopotential method freezes the core electrons and introduces smooth pseudowavefunctions and an effective one-body potential for the valence electrons, thereby greatly reducing the computational cost while preserving the essential content of all-electron calculations. In the following sections, we use pseudopotentials constructed in this manner to discuss concretely real-time electron dynamics calculations for molecules and solids.

## 4 Real-Time Electron Dynamics Calculations for Molecules

In the previous section, we introduced the pseudopotential method, which replaces the core electrons by an effective ionic potential and thereby makes it possible to treat valence Kohn–Sham orbitals efficiently. In this section, we use that framework to discuss real-time electron dynamics calculations for isolated molecules. Since a molecule is a finite system, it is placed in a sufficiently large computational box, and isolated boundary conditions are imposed so that the wavefunctions vanish at the boundary. Within this setting, we follow the time evolution of the electron density and the Kohn–Sham orbitals.

The purpose of this section is to understand, through example calculations using the first-principles code Octopus, which equations are solved in a real-time electron dynamics calculation and how response functions and spectra are extracted from the resulting time-dependent quantities. As a concrete example, we consider an isolated  $N_2$  molecule. For simplicity, spin degeneracy is assumed; namely, each spatial orbital is occupied by two electrons.

In what follows, the nuclear positions of the molecule are fixed. This corresponds to an electron dynamics calculation based on the Born–Oppenheimer approximation and is commonly used when one is interested in the short-time electronic response on time scales from sub-femtoseconds to several tens of femtoseconds. On longer time scales, or when nuclear motion and vibrational excitation are important, coupled electron–nuclear dynamics must be considered.

**What we learn in this section** In this section, we learn the basic workflow of real-time TDDFT calculations for molecules:

- First, we solve the ground-state Kohn–Sham equation and prepare the initial condition for time propagation.
- Next, we propagate the time-dependent Kohn–Sham equation in real time under a weak external electric field.
- From the resulting time-dependent density, we compute the dipole moment and evaluate the polarizability and optical absorption spectrum by Fourier analysis.
- Finally, we examine the meaning of Octopus input files, output files, and numerical parameters, and discuss convergence checks and physical interpretation of the results.

### 4.1 Ground-State Calculations for Molecules

As a preliminary step before performing a real-time electron dynamics calculation, one first obtains the molecular ground state. This ground-state calculation provides the initial condition for solving the time-dependent Kohn–Sham equation. It is therefore not merely a preparatory calculation: its accuracy directly affects the reliability of the time-dependent dipole moment and absorption spectrum obtained later.

#### 4.1.1 Kohn–Sham Equation for Atoms and Molecules

The ground state in the absence of external fields is obtained by solving the time-independent Kohn–Sham equation

$$\left[ \frac{\mathbf{p}^2}{2m} + \hat{v}_{\text{ion}} + v_{\text{H}}[\rho](\mathbf{r}) + v_{\text{xc}}[\rho](\mathbf{r}) \right] \phi_n(\mathbf{r}) = \epsilon_n \phi_n(\mathbf{r}) \quad (37)$$

self-consistently. Here,  $\phi_n(\mathbf{r})$  is a Kohn–Sham orbital,  $\epsilon_n$  is the corresponding one-particle energy, and  $\rho(\mathbf{r})$  is the electron density. The operator  $\hat{v}_{\text{ion}}$  is the pseudopotential describing the electron–ion interaction,  $v_{\text{H}}[\rho](\mathbf{r})$  is the Hartree potential, and  $v_{\text{xc}}[\rho](\mathbf{r})$  is the exchange–correlation potential. When pseudopotentials are used,  $\hat{v}_{\text{ion}}$  generally contains a nonlocal operator.

Assuming spin degeneracy, the electron density is constructed from the occupied Kohn–Sham orbitals as

$$\rho(\mathbf{r}) = 2 \sum_{n=1}^{N_{\text{occ}}} |\phi_n(\mathbf{r})|^2. \quad (38)$$

Here,  $N_{\text{occ}}$  is the number of occupied spatial orbitals, and the factor of 2 represents spin degeneracy. If the number of explicitly treated electrons is denoted by  $N_e$ , then

$$N_e = \int d\mathbf{r} \rho(\mathbf{r}) = 2N_{\text{occ}} \quad (39)$$

must be satisfied. When pseudopotentials are used,  $N_e$  is not the total number of electrons in the all-electron atom or molecule, but the number of valence electrons explicitly treated by the chosen pseudopotentials.

Although Eq. (37) looks like an ordinary eigenvalue problem, the Hartree and exchange-correlation potentials depend on the unknown density  $\rho(\mathbf{r})$ . Therefore, the orbitals cannot be obtained by a single diagonalization. In practice, one performs a self-consistent field (SCF) iteration:

1. Assume an initial density  $\rho^{(0)}(\mathbf{r})$ .
2. Construct the Kohn–Sham Hamiltonian  $\hat{h}_{\text{KS}}[\rho^{(i)}]$  from that density.
3. Solve the eigenvalue problem and obtain the orbitals  $\phi_n^{(i)}(\mathbf{r})$ .
4. Construct a new density  $\rho^{(i+1)}(\mathbf{r})$  using Eq. (38).
5. Repeat the procedure until the density, eigenvalues, or total energy changes sufficiently little.

When this iteration converges, the electron density and the effective potential are mutually consistent, and the resulting solution is regarded as the self-consistent ground state.

In a ground-state calculation, it is useful to check not only the total energy but also the energies of the highest occupied and lowest unoccupied orbitals. Absorption peaks obtained in a real-time calculation are sensitive to the ground-state electronic structure. If the ground state is not sufficiently converged, a reliable spectrum cannot be obtained even if the subsequent time propagation is performed.

#### 4.1.2 Ground-State Calculation Using Octopus

In the real-space implementation of Octopus, a molecule is placed on a real-space grid, and the Kohn–Sham equation is solved in a finite computational box. Source code 1 shows an example input file for a ground-state calculation of an  $N_2$  molecule. In this example, a ground-state calculation is specified as the calculation mode, and the isolated molecule is treated on a real-space grid.

[https://shunsuke-sato.github.io/page/etc/lecture\\_notes/src\\_tddft\\_note/lin\\_res\\_n2/inp\\_gs](https://shunsuke-sato.github.io/page/etc/lecture_notes/src_tddft_note/lin_res_n2/inp_gs)

Source Code 1: Octopus input file: ground state of an  $N_2$  molecule

```

1 CalculationMode = gs
2 FromScratch = yes
3
4 Radius = 10.0
5
6 XCFunctional = lda_x + lda_c_pw
7 Spacing = 0.4
8
9 ExtraStates = 2
10
11 PseudopotentialSet = hgh_lda

```

```

12
13 # User defined variable, (N-N distance)
14 NN_distance = 2.0
15
16 %Coordinates
17 "N" | 0 | 0 | 0.5*NN_distance
18 "N" | 0 | 0 | -0.5*NN_distance
19 %
20
21 Eigensolver = cg
22 MaximumIter = 60
23 ConvEigenError = yes
24 EigensolverTolerance = 1e-8

```

The main points needed to understand this input file are summarized below.

**CalculationMode = gs** This specifies a ground-state calculation. Before real-time propagation, this calculation is usually performed first.

**FromScratch = yes** This specifies that the calculation should start from scratch rather than reusing data produced in previous calculations. When performing convergence tests, one should be careful not to read old data unintentionally.

**Coordinates** This specifies the atomic species and positions that form the molecule. For an  $N_2$  molecule, the molecular axis and bond length are determined here. When an external field polarization direction is specified later, the relative orientation between the molecular axis and the polarization direction has physical meaning.

**Radius** This specifies the size of the computational region containing the isolated molecule. If the molecule is too close to the boundary, the tail of the electron density is artificially truncated, which can change the ground-state energy and excitation spectrum.

**Spacing** This specifies the spacing of the real-space grid. A smaller spacing improves accuracy but increases the number of grid points and therefore the computational cost.

**XCFUNCTIONAL** This specifies the exchange-correlation functional. Using the same approximation in the ground-state and time-dependent calculations keeps the initial state and time evolution theoretically consistent.

**PseudopotentialSet** This specifies the set of pseudopotentials to be used. Since the combination of pseudopotentials and exchange-correlation functional affects the accuracy of physical quantities, it should be checked for the Octopus version and pseudopotential database being used.

**Eigensolver, MaximumIter, EigensolverTolerance** These are related to the method for solving the eigenvalue problem, the maximum number of iterations, and convergence criteria. If the SCF calculation does not converge, one should reconsider the number of iterations, mixing method, initial density, and grid settings.

Source code 2 is an example shell script for running the ground-state calculation. In this example, the input file for the ground-state calculation is copied to the standard input-file name **inp** read by Octopus, and the standard output is saved to a log file. The path to the Octopus executable, the number of MPI processes, and the number of OpenMP threads depend on the computational environment and should be modified accordingly.

[https://shunsuke-sato.github.io/page/etc/lecture\\_notes/src\\_tddft\\_note/lin\\_res\\_n2/run\\_gs.sh](https://shunsuke-sato.github.io/page/etc/lecture_notes/src_tddft_note/lin_res_n2/run_gs.sh)

Source Code 2: Shell script for running the ground-state calculation of an  $N_2$  molecule

```

1 #!/bin/bash
2
3 export OMP_NUM_THREADS=2

```

```

4 export MY_OCTOPUS="/work/sato/octopus_tutorial/octopus_execute/install/bin/octopus"
5
6
7 cp inp_gs inp
8
9 mpirun -np 1 "${MY_OCTOPUS}" >log.gs.log 2>&1

```

After the calculation has finished, check the log file and confirm that the SCF iteration has converged properly. It is also important to vary **Radius** and **Spacing** and confirm that the ground-state energy and important eigenvalues change only negligibly. Such convergence checks are indispensable in calculations used for actual research.

## 4.2 Real-Time Propagation and Molecular Linear Response

Once the ground state has been obtained, the next step is to apply an external field and propagate the Kohn–Sham orbitals in time. If the external field is weak, the linear response function can be extracted from the resulting time-dependent dipole moment. This provides, as a time-domain initial-value problem, information equivalent to that obtained from frequency-domain linear-response TDDFT, such as excitation energies and oscillator strengths.

Here, a “real-time calculation” means that one does not compute excited states one by one in advance. Instead, the ground state is slightly perturbed by an external field, and the subsequent time signal is analyzed. This signal contains a superposition of the eigenmodes excited by the field. Therefore, by combining real-time propagation with Fourier analysis, one can investigate the response over a wide energy range in a single calculation.

### 4.2.1 Time-Dependent Kohn–Sham Equation

The wavelength of light is much longer than the typical size of a molecule. Therefore, the optical electric field acting on the molecule can be regarded as spatially uniform over the molecular region. This is called the long-wavelength approximation or the dipole approximation. For isolated molecules, this uniform electric field is often represented in the length gauge. In that case, the external potential acting on an electron can be written as  $-e\mathbf{E}(t) \cdot \mathbf{r}$ .

Under a spatially uniform external electric field  $\mathbf{E}(t)$ , the valence-electron dynamics of a molecule is described by the time-dependent Kohn–Sham equation

$$i\hbar \frac{\partial}{\partial t} \psi_m(\mathbf{r}, t) = \left[ \frac{\mathbf{p}^2}{2m} + \hat{v}_{\text{ion}} + v_{\text{H}}[\rho(\mathbf{r}, t)](\mathbf{r}, t) + v_{\text{xc}}[\rho(\mathbf{r}, t)](\mathbf{r}, t) - e\mathbf{E}(t) \cdot \mathbf{r} \right] \psi_m(\mathbf{r}, t). \quad (40)$$

Here,  $\psi_m(\mathbf{r}, t)$  is a time-dependent Kohn–Sham orbital and coincides with the ground-state orbital  $\phi_m(\mathbf{r})$  at the initial time. The time-dependent electron density is given by

$$\rho(\mathbf{r}, t) = 2 \sum_{m=\text{occ}} |\psi_m(\mathbf{r}, t)|^2. \quad (41)$$

The factor of 2 again represents spin degeneracy.

An important point in Eq. (40) is that the electron density  $\rho(\mathbf{r}, t)$  changes with time, and the Hartree and exchange-correlation potentials are updated accordingly. Thus, the electrons are not merely driven independently by the external field; they also feel the response of the other electrons through the density-dependent change of the mean field.

The exact exchange-correlation potential may depend on the density at all previous times. This property is called a memory effect. In many practical calculations, however, one uses an adiabatic approximation, in which the exchange-correlation potential at each time is approximated as a functional of only the instantaneous density. Typical examples are obtained by applying ground-state LDA or GGA functionals to the time-dependent density at each time. The adiabatic approximation is widely used, but one should keep in mind its limitations in problems where double excitations, long-time relaxation, or strong electron correlation are important.



#### 4.2.2 Dipole Moment, Current, and Optical Absorption Spectrum

The basic quantity characterizing the optical response of a molecule is the time-dependent electric dipole moment. We define the induced dipole moment relative to the ground state as

$$\mathbf{d}(t) = -e \int d\mathbf{r} \, \mathbf{r} [\rho(\mathbf{r}, t) - \rho(\mathbf{r}, t_0)]. \quad (42)$$

Here,  $\rho(\mathbf{r}, t_0)$  is the ground-state density before the external field is applied. Since the ground-state dipole moment is subtracted in Eq. (42),  $\mathbf{d}(t)$  represents the electronic dipole moment induced by the external field. For the homonuclear diatomic molecule  $N_2$  with fixed nuclei, the nuclear contribution is time-independent, so Eq. (42) is sufficient when discussing the induced response.

When the external electric field is sufficiently weak, the induced dipole moment responds linearly to the field. In that case,

$$d_a(t) = \sum_{b=\{x,y,z\}} \int_{-\infty}^{\infty} dt' \alpha_{ab}(t-t') E_b(t') \quad (43)$$

can be written. Here,  $d_a(t)$  and  $E_b(t)$  are the  $a$  component of  $\mathbf{d}(t)$  and the  $b$  component of  $\mathbf{E}(t)$ , respectively, and  $\alpha_{ab}(t)$  is the polarizability tensor in the time domain. Since we consider linear response from the ground state and the system has time-translation symmetry, the response function depends only on the time difference  $t - t'$ , not on the two times separately.

Fourier transforming Eq. (43) gives

$$\tilde{d}_a(\omega) = \sum_{b=\{x,y,z\}} \tilde{\alpha}_{ab}(\omega) \tilde{E}_b(\omega). \quad (44)$$

In these notes, the Fourier transforms are defined as

$$\tilde{d}_a(\omega) = \int_{-\infty}^{\infty} dt \, d_a(t) e^{i\omega t}, \quad (45)$$

$$\tilde{\alpha}_{ab}(\omega) = \int_{-\infty}^{\infty} dt \, \alpha_{ab}(t) e^{i\omega t}, \quad (46)$$

$$\tilde{E}_b(\omega) = \int_{-\infty}^{\infty} dt \, E_b(t) e^{i\omega t}. \quad (47)$$

Let us now recall an important property of the polarizability tensor  $\tilde{\alpha}_{ab}(\omega)$ . Although the electric dipole moment in Eq. (43) is driven by the electric field  $E(t)$ , causality requires that an electric field at a future time cannot affect the dipole moment at the present time. In other words, the electric field  $\mathbf{E}(t')$  at a future time ( $t' > t$ ) has no influence on  $\mathbf{d}(t)$ . This is guaranteed by the fact that the time-domain polarizability tensor  $\alpha_{ab}(t)$  vanishes for  $t < 0$ . Therefore,

$$\alpha_{ab}(t) = \Theta(t) \alpha_{ab}(t) \quad (48)$$

holds, where  $\Theta(t)$  is the Heaviside step function. Fourier transforming both sides gives

$$\tilde{\alpha}_{ab}(\omega) = \frac{1}{2\pi} \int_{-\infty}^{\infty} d\omega' \, \tilde{\Theta}(\omega - \omega') \tilde{\alpha}_{ab}(\omega'). \quad (49)$$

Here,  $\tilde{\Theta}(\omega)$  is the Fourier transform of  $\Theta(t)$  and is given by

$$\begin{aligned} \tilde{\Theta}(\omega) &= \lim_{\gamma \rightarrow 0} \int_{-\infty}^{\infty} dt \, \Theta(t) e^{i\omega t} e^{-|\gamma|t} \\ &= \lim_{\gamma \rightarrow 0} \frac{i}{\omega + i|\gamma|} \\ &= \pi \delta(\omega) + i\mathcal{P} \frac{1}{\omega}. \end{aligned} \quad (50)$$

Here,  $\mathcal{P}$  denotes the Cauchy principal value. Using this expression, Eq. (49) becomes

$$\begin{aligned}\tilde{\alpha}_{ab}(\omega) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} d\omega' \left[ \pi \delta(\omega - \omega') + i\mathcal{P} \frac{1}{\omega - \omega'} \right] \tilde{\alpha}_{ab}(\omega') \\ &= \frac{1}{2} \tilde{\alpha}_{ab}(\omega) + \frac{i}{2\pi} \mathcal{P} \int_{-\infty}^{\infty} d\omega' \frac{\tilde{\alpha}_{ab}(\omega')}{\omega - \omega'}.\end{aligned}\quad (51)$$

Thus,

$$\tilde{\alpha}_{ab}(\omega) = \frac{i}{\pi} \mathcal{P} \int_{-\infty}^{\infty} d\omega' \frac{\tilde{\alpha}_{ab}(\omega')}{\omega - \omega'} \quad (52)$$

is obtained. Considering the real and imaginary parts of both sides gives the **Kramers–Kronig relations**

$$\text{Re} [\tilde{\alpha}_{ab}(\omega)] = \frac{1}{\pi} \mathcal{P} \int_{-\infty}^{\infty} d\omega' \frac{\text{Im} [\tilde{\alpha}_{ab}(\omega')]}{\omega' - \omega}, \quad (53)$$

$$\text{Im} [\tilde{\alpha}_{ab}(\omega)] = -\frac{1}{\pi} \mathcal{P} \int_{-\infty}^{\infty} d\omega' \frac{\text{Re} [\tilde{\alpha}_{ab}(\omega')]}{\omega' - \omega}. \quad (54)$$

In this way, causality relates the real and imaginary parts of the Fourier transform of a response function.

The optical absorption spectrum of a molecule is closely related to the imaginary part of the polarizability. For example, in a convention often used in atomic units, the orientationally averaged polarizability is defined as

$$\bar{\alpha}(\omega) = \frac{1}{3} [\alpha_{xx}(\omega) + \alpha_{yy}(\omega) + \alpha_{zz}(\omega)], \quad (55)$$

and the absorption cross section is written as

$$\sigma_{\text{abs}}(\omega) = \frac{4\pi\omega}{c} \text{Im} \bar{\alpha}(\omega). \quad (56)$$

Here,  $c$  is the speed of light. The prefactor depends on the unit system and Fourier-transform convention, so one must always check the units used in an actual analysis script. If only peak positions and relative intensities are discussed,  $\text{Im} \alpha_{aa}(\omega)$  itself is often plotted as a quantity corresponding to the absorption spectrum.

The peaks appearing in the imaginary part of the polarizability correspond to electronic excitations that are allowed as electric-dipole transitions for the polarization direction of the external field. The peak positions mainly reflect excitation energies, while the peak intensities reflect the magnitude of transition dipole moments. However, Kohn–Sham eigenvalue differences themselves do not generally give the excitation energies. Peaks obtained from real-time electron dynamics should be interpreted as the response including electron–electron interaction effects mediated by the time-dependent Hartree and exchange–correlation potentials.

#### 4.2.3 Extracting Linear Response from an Impulsive Perturbation

In practical real-time linear-response calculations, it is convenient to use a weak delta-function-like electric field. Consider, for example, an impulsive electric field of the form

$$\mathbf{E}(t) = k_0 \mathbf{e}_b \delta(t). \quad (57)$$

Here,  $k_0$  is the strength of the perturbation, and  $\mathbf{e}_b$  is the unit vector in the  $b$  direction. Substituting Eq. (57) into Eq. (43) gives

$$d_a(t) = k_0 \alpha_{ab}(t). \quad (58)$$

Therefore, by Fourier transforming the dipole moment induced by a weak delta-like perturbation, the polarizability tensor can be obtained as

$$\tilde{\alpha}_{ab}(\omega) = \frac{\tilde{d}_a(\omega)}{k_0}. \quad (59)$$

The advantage of this method is that a broad frequency range of the response can be obtained from a single real-time propagation. An ideal delta-function perturbation contains all frequency components, so the time-dependent dipole moment contains information on many excited modes of the system.

One should note, however, that Eq. (59) is valid only in the linear-response regime. If the perturbation amplitude is too large, second- and higher-order nonlinear responses are mixed in, and the polarizability cannot be extracted correctly. In practical calculations, it is good practice to check that the resulting  $\alpha(\omega)$  changes negligibly when the perturbation strength is reduced, for example, by a factor of two.

### 4.3 Real-Time Linear-Response Calculation Using Octopus

In this subsection, we summarize the concrete workflow for an  $N_2$  molecule, from the ground-state calculation to real-time propagation and spectral analysis. In Octopus, many settings are written in an input file named `inp`, and ground-state and time-dependent calculations are performed by changing the calculation mode.

#### 4.3.1 Workflow

A typical real-time linear-response calculation for a molecule proceeds as follows.

1. Choose the molecular structure, computational box size, grid spacing, exchange-correlation functional, and pseudopotentials.
2. Run a ground-state calculation with `CalculationMode = gs` and confirm SCF convergence.
3. Using the same structure, grid, functional, and pseudopotentials as in the ground-state calculation, run a time-dependent calculation with `CalculationMode = td`.
4. Apply a weak delta-function-like perturbation and examine the time-dependent dipole moment written, for example, to `td.general/multipoles`.
5. Fourier transform the dipole moment and obtain the polarizability using Eq. (59).
6. Change `Spacing`, `Radius`, `TDTimeStep`, `TDPropagationTime`, and `TDDeltaStrength`, and check the convergence of peak positions and intensities.

The important point in this workflow is not to regard the time-dependent calculation in isolation. Only when the ground-state convergence, field strength, stability of the time propagation, and Fourier-analysis conditions are all properly controlled can the spectrum be interpreted as a physical quantity.

#### 4.3.2 Input File for Real-Time Propagation

Source code 3 shows an Octopus input file for applying a weak delta-function-like perturbation to an  $N_2$  molecule and calculating the subsequent electron dynamics. Since the actual input-variable names and available values may depend on the Octopus version, one should consult the manual for the version being used.

[https://shunsuke-sato.github.io/page/etc/lecture\\_notes/src\\_tddft\\_note/lin\\_res\\_n2/inp\\_td](https://shunsuke-sato.github.io/page/etc/lecture_notes/src_tddft_note/lin_res_n2/inp_td)

Source Code 3: Octopus input file: linear response of an  $N_2$  molecule

```

1 CalculationMode = td
2 FromScratch = yes
3
4 Radius = 10.0
5
6 XCFunctional = lda_x + lda_c_pw
7 Spacing = 0.4
8
9 ExtraStates = 0
10
11 PseudopotentialSet = hgh_lda
12
13 # User defined variable, (N-N distance)
14 NN_distance = 2.0
15
16 %Coordinates
17 "N" | 0 | 0 | 0.5*NN_distance
18 "N" | 0 | 0 | -0.5*NN_distance
19 %
20
21 Eigensolver = cg
22 MaximumIter = 60
23 ConvEigError = yes
24 EigensolverTolerance = 1e-8
25
26
27 TDPropagator = aetrs
28 TDExponentialMethod = taylor
29 TDExpOrder = 4
30 TDPropagationTime = 10*fs
31 TDTimeStep = 0.08
32
33 TDDeltaStrength = 0.01
34 TDPolarizationDirection = 3

```

In this input file, the same molecular structure, grid, exchange-correlation functional, and pseudopotentials as in the ground-state calculation are used, and settings related to time propagation are added. The main input variables are as follows.

**CalculationMode** This specifies the mode for a time-dependent calculation. The orbitals and density obtained from the ground-state calculation are read as the initial state, and time propagation is then started.

**TDPropagator** This specifies the approximation used for the time-evolution operator. The value **aetrs** shown in the example is often used as a real-time propagator that takes time-reversal symmetry into account.

**TDExponentialMethod, TDExpOrder** These specify how the short-time time-evolution operator is evaluated. If a Taylor expansion is used, increasing the expansion order improves accuracy but also increases the computational cost.

**TDPropagationTime** This specifies the total propagation time. A longer propagation time improves the frequency resolution, but it also increases the computational time.

**TDTimeStep** This specifies the time step. If the time step is too large, numerical errors in the time propagation increase and can affect energy conservation and high-frequency components of the spectrum.

**TDDeltaStrength** This specifies the strength of the delta-like perturbation. To obtain a linear response, this strength must be sufficiently small.

**TDPolarizationDirection** This specifies the polarization direction of the delta-like perturbation. The resulting component of the polarizability depends on the relation between this polarization direction and the molecular axis of  $N_2$ .

Octopus uses atomic units internally. On the other hand, in input files it may be possible to specify units such as `fs` and `angstrom` explicitly. In atomic units, the unit of time is  $\hbar/E_h$ , and the unit of energy is the Hartree. For example, when a time step is given in atomic units, it is useful to keep its conversion to femtoseconds in mind when discussing the result with researchers in other fields. Therefore, attention to unit conversion is always important.

### 4.3.3 Running the Calculation

Source code 4 shows a shell script for running the time-dependent calculation using the input file in source code 3. As in the ground-state calculation, the path to the Octopus executable, the number of MPI processes, and the number of OpenMP threads should be adjusted according to each computational environment.

[https://shunsuke-sato.github.io/page/etc/lecture\\_notes/src\\_tddft\\_note/lin\\_res\\_n2/run\\_td.sh](https://shunsuke-sato.github.io/page/etc/lecture_notes/src_tddft_note/lin_res_n2/run_td.sh)

Source Code 4: Shell script for running the linear-response calculation of an  $N_2$  molecule

```

1 #!/bin/bash
2
3 export OMP_NUM_THREADS=4
4 export MY_OCTOPUS="/work/sato/octopus_tutorial/octopus_execute/install/bin/octopus"
5
6 cp inp_td inp
7
8 mpirun -np 2 "${MY_OCTOPUS}" >log_td.log 2>&1

```

When the time-dependent calculation finishes normally, the time evolution of the dipole and multipole moments is written to output files. In the analysis below, we use the time signal of the dipole moment stored in `td.general/multipoles`.

After the calculation, it is useful to check at least the following files and directories.

**log\_td.log** This is the log file containing the standard output of the time-dependent calculation.

Check whether the time propagation reached the final time and whether warnings or errors were reported.

**td.general/multipoles** This file stores the time evolution of dipole and multipole moments.

Since the meaning of the columns may depend on the Octopus version and output settings, one should check the comment lines at the beginning of the file and the corresponding manual.

**td.general/energy** Information on the energy during the time-dependent calculation may be written here. By checking whether there is an unnatural drift after the external field has ended, one can assess the validity of the time step and propagator.

**restart** This directory stores data used to restart ground-state or time-dependent calculations.

When performing calculations under different conditions, care is needed with the `FromScratch` setting to avoid unintended reuse of old data.

The time evolution of the dipole moment obtained from the calculation above is shown in Fig. 1.

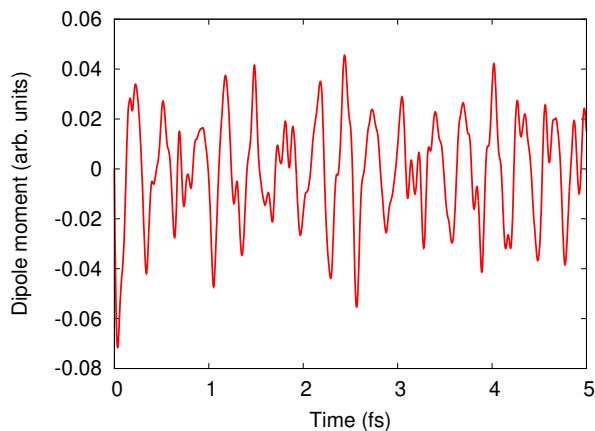


Figure 1: Time evolution of the dipole moment

In Fig. 1, the dipole moment oscillates after the delta-like perturbation. This oscillation reflects the fact that the electronic system of the molecule evolves as a superposition of several excited states. By analyzing the frequency components of this oscillation, peaks corresponding to allowed transition energies can be obtained.

## 4.4 Spectral Analysis and Choice of Numerical Parameters

### 4.4.1 Fourier Analysis of the Dipole Moment

Source code 5 shows a Python script for analyzing the dipole moment in Fig. 1 and evaluating the molecular polarizability tensor.

[https://shunsuke-sato.github.io/page/etc/lecture\\_notes/src\\_tddft\\_note/lin\\_res\\_n2/optical\\_response.py](https://shunsuke-sato.github.io/page/etc/lecture_notes/src_tddft_note/lin_res_n2/optical_response.py)

Source Code 5: Python script for evaluating the polarizability of an  $N_2$  molecule

```

1  # /// script
2  # dependencies = [
3  #   "pandas",
4  #   "matplotlib",
5  #   "numpy"
6  # ]
7  # ///
8
9  import numpy as np
10 import pandas as pd
11 import matplotlib.pyplot as plt
12
13
14 def main():
15     # Input file
16     filename = "td.general/multipoles"
17
18     # Read data (ignore lines starting with '#')
19     df = pd.read_csv(
20         filename,
21         comment="#",
22         sep=r"\s+",
23         header=None,
24         engine="python"
25     )
26
27     # Assign column names
28     df.columns = [
29         "Iter",
30         "t",
31         "Electronic_charge",
32         "x",
33         "y",

```

```

34     "z",
35 ]
36
37 # Show basic info
38 print("Data preview:")
39 print(df.head())
40 print("\nData shape:", df.shape)
41
42 # Extract columns as NumPy arrays
43 tt = df["t"].to_numpy()
44 xt = df["x"].to_numpy()
45 yt = df["y"].to_numpy()
46 zt = df["z"].to_numpy()
47
48 dt = tt[1]-tt[0]
49
50 kick_k = 0.01
51 kick_dir = np.zeros(3)
52 kick_dir[0] = 0.0
53 kick_dir[1] = 0.0
54 kick_dir[2] = 1.0
55
56 ev = 27.2114
57 nomega = 400
58 omega_ini = 0.1 / ev
59 omega_fin = 20.0 / ev
60 gamma = 0.5 / ev # damping factor; adjust as needed
61
62 omega = np.linspace(omega_ini, omega_fin, nomega)
63 domega = omega[1] - omega[0]
64
65 alpha = np.zeros(nomega, dtype=complex)
66
67 for iw in range(nomega):
68     xw = 0.0 + 0.0j
69     yw = 0.0 + 0.0j
70     zw = 0.0 + 0.0j
71
72     for it in range(len(tt)):
73         phase = np.exp((1j * omega[iw] - gamma) * tt[it])
74         xw += phase * xt[it]
75         yw += phase * yt[it]
76         zw += phase * zt[it]
77
78     alpha[iw] = -dt*(
79         kick_dir[0] * xw +
80         kick_dir[1] * yw +
81         kick_dir[2] * zw
82     ) / kick_k
83
84 # Example output
85 print("\nFirst 10 omega values:")
86 print(omega[:10])
87
88 print("\nFirst 10 alpha values:")
89 print(alpha[:10])
90
91 # Optional plot
92 plt.figure()
93 plt.plot(omega * ev, alpha.real, label="Re(alpha)")
94 plt.plot(omega * ev, alpha.imag, label="Im(alpha)")
95 plt.xlim(0.0, 20.0)
96 plt.xlabel("Omega (eV)")
97 plt.ylabel("Alpha")
98 plt.legend()
99 plt.tight_layout()
100 # plt.show()
101 plt.savefig("polarizability.png")
102
103
104 if __name__ == "__main__":
105     main()

```

In a finite-time numerical calculation, the Fourier-transform integral is restricted to the interval  $0 < t < T_{\text{prop}}$ . Therefore, one often performs the Fourier transform with an exponential damping

factor  $e^{-\gamma t}$  as

$$\tilde{d}_a^\gamma(\omega) = \int_0^{T_{\text{prop}}} dt [d_a(t) - d_a(0)] e^{i\omega t} e^{-\gamma t}. \quad (60)$$

Here,  $\gamma$  is a parameter controlling the damping width. Since energy and angular frequency are expressed in the same units in atomic units, if  $\gamma$  is specified in eV, it must be converted to Hartree.

Multiplying by the damping factor corresponds to introducing a finite linewidth into the spectrum. If  $\gamma$  is made large, the spectrum becomes smoother, but nearby peaks become harder to resolve. If  $\gamma$  is made small, the peaks become sharper, but oscillations due to finite-time truncation become more noticeable.

The total propagation time  $T_{\text{prop}}$  determines the spectral resolution. The approximate energy resolution can be estimated as

$$\Delta E \simeq \frac{2\pi\hbar}{T_{\text{prop}}}. \quad (61)$$

For example, for  $T_{\text{prop}} = 10$  fs,  $\Delta E$  is roughly 0.4 eV. If one wishes to discuss finer peak structures, a longer propagation time is required.

The time step  $\Delta t$  is related not only to the stability of the time propagation but also to the upper limit of the high-frequency region that can be analyzed. For a sampled time signal, frequency components above the Nyquist frequency, roughly  $\omega_{\text{max}} \sim \pi/\Delta t$ , cannot be treated reliably. Thus, if high-energy absorption peaks are of interest, a sufficiently small `TDTimeStep` must be chosen.

The imaginary part of the polarizability of the  $N_2$  molecule evaluated using source code 5 is shown in Fig. 2.

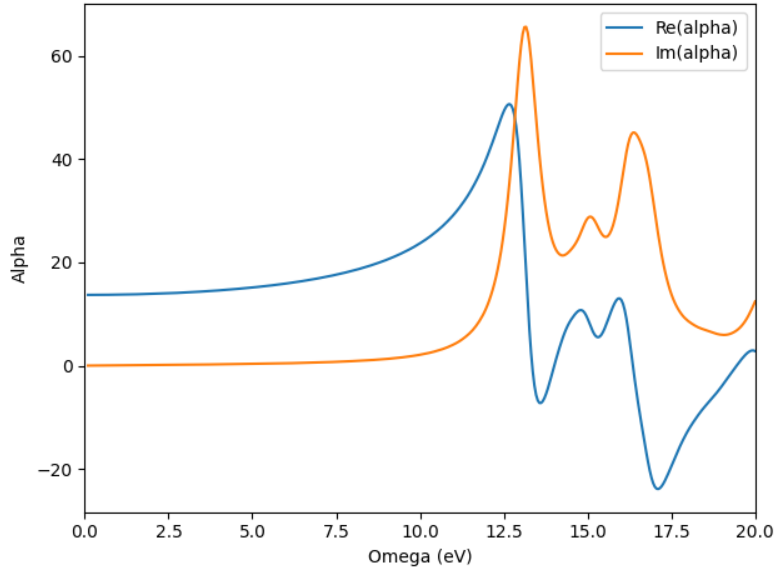


Figure 2: Polarizability

The peaks in Fig. 2 correspond to electronic excitations of the  $N_2$  molecule. The peak positions mainly reflect excitation energies, and the peak intensities reflect the size of the transition dipole moment for the polarization direction of the external field. However, peak positions and intensities obtained from real-time TDDFT depend on the exchange-correlation functional, pseudopotentials, grid, computational box size, and time-propagation parameters. Therefore, convergence checks with respect to the computational conditions are indispensable.



#### 4.4.2 Choice of Numerical Parameters

In real-time TDDFT calculations for molecules, the following numerical parameters are particularly important.

**Grid spacing** `Spacing` If the grid is too coarse, the spatial variation of the Kohn–Sham orbitals cannot be represented accurately, and the ground-state energy, eigenvalues, dipole moment, and spectral peaks may change. One should first check convergence of the ground-state energy and, when necessary, also check convergence of the spectrum.

**Computational box size** `Radius` If the box is too small, the tail of the electron density and the spatial extent of excited states are cut off at the boundary. A large box is especially important for Rydberg-type excitations or strong-field responses involving ionization.

**Time step** `TDTimeStep` If the time step is too large, the approximation error in the time-evolution operator increases. To analyze high excitation energies, a smaller time step is required. It is useful to confirm that the dipole moment and spectrum change only negligibly when the time step is reduced, for example, by a factor of two.

**Total propagation time** `TDPropagationTime` The total propagation time determines the frequency resolution. When discussing peak widths or closely spaced peaks, one should choose a sufficiently long propagation time, using Eq. (61) as a guide.

**Perturbation strength** `TDDeltaStrength` To obtain a linear response, the perturbation must be sufficiently weak. One should choose a range in which the polarizability does not change when the perturbation strength is varied. If the perturbation is too strong, second- and higher-order responses are mixed in.

**Damping factor**  $\gamma$  This is the artificial linewidth introduced in the Fourier analysis. If it is too large, peaks are excessively broadened; if it is too small, oscillations due to finite-time effects remain.

**Absorbing boundary** For bound-state linear response induced by a weak delta-like perturbation, calculations without an absorbing boundary are often sufficient if the box is large enough. In contrast, when ionization occurs under a strong laser field or when an electron wave packet is emitted into continuum states, an absorbing boundary is needed to prevent wave packets reflected at the box boundary from returning to the molecular region. The specific input-variable names and usage for absorbing boundaries depend on the Octopus version, so the manual for the computational environment being used should be checked.

**Number of empty orbitals** `ExtraStates` In ground-state calculations, unoccupied orbitals may be computed for analysis or visualization in addition to occupied orbitals. However, real-time propagation itself constructs the density from the time evolution of occupied orbitals, so the role of `ExtraStates` should be understood in relation to the calculation mode and output purpose.

In actual numerical calculations, convergence should not be judged from a single parameter alone. For example, even if the ground-state energy is converged, the high-energy region of the excitation spectrum may still be sensitive to the grid spacing. Also, changing the computational box size can significantly affect peaks associated with Rydberg states or excitations close to the continuum. For research calculations, it is desirable to check convergence at least with respect to `Spacing`, `Radius`, `TDTimeStep`, and `TDPropagationTime`.

#### 4.4.3 Common Pitfalls and Physical Interpretation

Finally, we summarize common pitfalls in real-time TDDFT calculations for molecules.

**Unconverged ground state** If time propagation is started from an insufficiently converged SCF solution, oscillations originating from numerical inconsistency in the initial state may be mixed into the response to the external field. Before starting the time-dependent calculation, one should confirm that the ground-state energy and density are sufficiently converged.

**Perturbation too strong** When linear response is desired, the dipole moment must be proportional to the perturbation strength. One should check that  $\tilde{d}(\omega)/k_0$  does not change when `TDDeltaStrength` is varied.

**Computational box too small** If the tail of the electron density or the wavefunction of an excited state is cut off at the boundary, artificial confinement shifts the peak positions. A particularly large computational box is needed for Rydberg excitations or ionization.

**Propagation time too short** If the total propagation time is too short, nearby peaks cannot be resolved. The spectral resolution can be estimated using Eq. (61), so `TDPropagationTime` should be chosen according to the desired energy resolution.

**Interpretation of the damping width** The parameter  $\gamma$  introduced in Fourier analysis can serve both as numerical smoothing and as a simple model of finite lifetime. However, the linewidth introduced here does not necessarily represent an actual relaxation time. In a closed-electron-system calculation within adiabatic TDDFT, irreversible relaxation due to electron–electron scattering or electron–phonon scattering is usually not included explicitly.

**Limitations of the exchange-correlation functional** Adiabatic LDA or GGA often gives qualitatively useful optical responses, but errors can become large for charge-transfer excitations, Rydberg excitations, double excitations, or situations where long-range exchange-correlation effects are important. When comparing with experiment or high-accuracy calculations, one should always keep the functional dependence in mind.

Taking these points together, a molecular real-time TDDFT calculation is not simply a matter of running an input file and plotting a spectrum. It is a calculation in which the ground state, external field, time propagation, Fourier analysis, and convergence checks must be designed consistently as one workflow. In the next section, we extend this idea to solids with periodic boundary conditions, where optical responses are evaluated from current response rather than from the dipole moment.

## 5 Real-Time Electron Dynamics Calculations for Solids

In the previous section, we discussed real-time electron dynamics calculations for isolated systems such as molecules. For isolated systems, the electric dipole moment induced by an external electric field can be computed directly, and polarizabilities and optical absorption spectra can be obtained from its Fourier analysis. In solids, however, periodic boundary conditions are imposed, and one must be careful in using the position operator  $\mathbf{r}$  and the electric dipole moment. In this section, we introduce the basic formulation of real-time electron dynamics calculations for periodic systems and explain how optical conductivity, dielectric functions, transient optical responses, and high-harmonic spectra can be evaluated from the current response.

The main differences between molecular and solid-state calculations are the following:

- Electronic states are represented by Bloch orbitals, and calculations are performed for  $\mathbf{k}$  points in the Brillouin zone.
- In order to describe a spatially uniform electric field in a way compatible with periodic boundary conditions, the velocity gauge is often used instead of the length gauge.
- Optical responses are evaluated from the unit-cell-averaged current density  $\mathbf{J}(t)$  rather than from the dipole moment.

We explain these points in order below.

**What we learn in this section** In this section, we learn the following points:

- how to formulate the ground-state Kohn–Sham equation and Bloch orbitals under periodic boundary conditions;
- why a uniform electric field is introduced through a vector potential in the velocity gauge;
- how gauge factors and current operators should be treated when nonlocal pseudopotentials are used;
- how to obtain the optical conductivity  $\sigma(\omega)$  and dielectric function  $\epsilon(\omega)$  from the current computed in real-time simulations;
- how pump–probe calculations can be simulated numerically to extract transient optical properties of nonequilibrium solids;
- how to organize ground-state, linear-response, and high-harmonic calculations for solids using Octopus;
- how to check convergence with respect to grid spacing, time step, propagation time, and  $\mathbf{k}$ -point sampling.

### 5.1 Ground-State Calculations for Solids

In this subsection, we review the ground-state calculation required as the initial condition for real-time simulations of solids. As in molecular calculations, the initial state for real-time electron dynamics is usually taken to be the ground state before the external field is applied. The essential difference is that, for solids, periodic boundary conditions are imposed and the electronic states are sampled over the Brillouin zone.

### 5.1.1 Kohn–Sham Equation for Solids

Consider a crystal in which ions are arranged periodically. Let  $\mathbf{a}_i$  be the primitive lattice vectors, and let a general lattice vector be written as

$$\mathbf{R} = n_1 \mathbf{a}_1 + n_2 \mathbf{a}_2 + n_3 \mathbf{a}_3. \quad (62)$$

The ionic potential then satisfies the periodicity condition

$$\hat{v}_{\text{ion}}(\mathbf{r} + \mathbf{R}) = \hat{v}_{\text{ion}}(\mathbf{r}). \quad (63)$$

In the ground state, the electron density  $\rho(\mathbf{r})$  is also expected to have the same lattice periodicity. Therefore, the Hartree potential  $v_{\text{H}}[\rho](\mathbf{r})$  and the exchange-correlation potential  $v_{\text{xc}}[\rho](\mathbf{r})$  are also lattice-periodic. The time-independent Kohn–Sham equation is then written as

$$\left[ \frac{\mathbf{p}^2}{2m} + \hat{v}_{\text{ion}} + v_{\text{H}}[\rho](\mathbf{r}) + v_{\text{xc}}[\rho](\mathbf{r}) \right] \phi_n(\mathbf{r}) = \epsilon_n \phi_n(\mathbf{r}). \quad (64)$$

Here,  $\mathbf{p} = -i\hbar\nabla$  is the momentum operator,  $\hat{v}_{\text{ion}}$  is the ionic potential including pseudopotentials,  $v_{\text{H}}$  is the Hartree potential, and  $v_{\text{xc}}$  is the exchange-correlation potential.

Because the Kohn–Sham Hamiltonian has the same periodicity as the crystal, Bloch’s theorem allows the Kohn–Sham orbitals to be written as

$$\phi_{b\mathbf{k}}(\mathbf{r}) = \exp[i\mathbf{k} \cdot \mathbf{r}] u_{b\mathbf{k}}(\mathbf{r}). \quad (65)$$

Here,  $b$  is the band index,  $\mathbf{k}$  is the Bloch wave vector, and  $u_{b\mathbf{k}}(\mathbf{r})$  is a lattice-periodic function satisfying

$$u_{b\mathbf{k}}(\mathbf{r} + \mathbf{R}) = u_{b\mathbf{k}}(\mathbf{r}). \quad (66)$$

Substituting Eq. (65) into Eq. (64), one obtains the Kohn–Sham equation for the periodic part  $u_{b\mathbf{k}}(\mathbf{r})$ :

$$\left[ \frac{(\mathbf{p} + \hbar\mathbf{k})^2}{2m} + e^{-i\mathbf{k} \cdot \mathbf{r}} \hat{v}_{\text{ion}} e^{i\mathbf{k} \cdot \mathbf{r}} + v_{\text{H}}[\rho](\mathbf{r}) + v_{\text{xc}}[\rho](\mathbf{r}) \right] u_{b\mathbf{k}}(\mathbf{r}) = \epsilon_{b\mathbf{k}} u_{b\mathbf{k}}(\mathbf{r}). \quad (67)$$

If the ionic potential is local, the factor  $e^{-i\mathbf{k} \cdot \mathbf{r}} \hat{v}_{\text{ion}} e^{i\mathbf{k} \cdot \mathbf{r}}$  is simply equal to  $\hat{v}_{\text{ion}}$ . However, if the pseudopotential contains nonlocal terms, this phase factor gives a nontrivial contribution. This point is also important when an external electric field is introduced in the velocity gauge.

The electron density is evaluated from the occupied Bloch orbitals as

$$\rho(\mathbf{r}) = \frac{1}{N_k} \sum_{\mathbf{k}} \sum_b f_{b\mathbf{k}} |u_{b\mathbf{k}}(\mathbf{r})|^2. \quad (68)$$

Here,  $N_k$  is the number of  $\mathbf{k}$  points, and  $f_{b\mathbf{k}}$  is the occupation factor. If spin degeneracy is not treated explicitly, the factor associated with spin degeneracy is included in  $f_{b\mathbf{k}}$ .

In real-time simulations of solids, the convergence of the ground-state band structure, band gap, electron density, and  $\mathbf{k}$ -point sampling directly affects the resulting optical response. Even if the total energy appears to be converged, the dielectric function or high-harmonic spectrum may still be sensitive to the numerical parameters. Therefore, convergence should be checked for the physical quantity that will ultimately be discussed.

### 5.1.2 Ground-State Calculation Using Octopus

Source code 6 shows an example Octopus input file for a ground-state calculation of bulk *Si*. Since the names and usage of input variables may depend on the version of Octopus, one should also consult the manual of the version being used.

[https://shunsuke-sato.github.io/page/etc/lecture\\_notes/src\\_tddft\\_note/lin\\_res\\_si/inp\\_gs](https://shunsuke-sato.github.io/page/etc/lecture_notes/src_tddft_note/lin_res_si/inp_gs)

Source Code 6: Octopus input file: ground state of bulk *Si*

```

1  CalculationMode = gs
2
3  PeriodicDimensions = 3
4
5  XCFunctional = lda_x + lda_c_pw
6
7
8
9  ExtraStates = 4
10
11 a = 10.18
12 %LatticeParameters
13   a | a | a
14 %
15
16 Spacing = (a/sqrt(2.0))/16.1
17
18 %LatticeVectors
19   0.0 | 0.5 | 0.5
20   0.5 | 0.0 | 0.5
21   0.5 | 0.5 | 0.0
22 %
23
24 %ReducedCoordinates
25   "Si" | 0.0 | 0.0 | 0.0
26   "Si" | 1/4 | 1/4 | 1/4
27 %
28
29 nk = 5
30 %KPointsGrid
31   nk | nk | nk
32 %
33
34 Eigensolver = cg
35 MaximumIter = 60
36 ConvEigenError = yes
37 EigensolverTolerance = 1e-8
38
39 KPointsUseSymmetries = yes
40 %SymmetryBreakDir
41   1 | 0 | 0
42 %
43
44 %Output
45   dos
46 %

```

[https://shunsuke-sato.github.io/page/etc/lecture\\_notes/src\\_tddft\\_note/lin\\_res\\_si/run\\_gs.sh](https://shunsuke-sato.github.io/page/etc/lecture_notes/src_tddft_note/lin_res_si/run_gs.sh)

Source Code 7: Shell script for running the ground-state calculation of bulk *Si*

```

1  #!/bin/bash
2
3  export OMP_NUM_THREADS=2
4  export MY_OCTOPUS="/work/sato/octopus_tutorial/octopus_execute/install/bin/octopus"
5
6
7  cp inp_gs inp
8
9  mpirun -np 4 "${MY_OCTOPUS}" >log.gs.log 2>&1

```

The important input items for ground-state calculations of solids are the lattice vectors, atomic coordinates, periodic boundary conditions, real-space grid,  $\mathbf{k}$ -point sampling, exchange-correlation functional, and pseudopotentials. In molecular calculations, one specifies a finite box containing an isolated molecule. In solid-state calculations, by contrast, the unit cell is repeated periodically, and the specification of lattice vectors and periodic boundary conditions becomes essential.

A typical workflow is as follows:

1. Determine the crystal structure and specify the lattice vectors and atomic coordinates.
2. Set periodic boundary conditions and specify the real-space grid spacing and  $\mathbf{k}$ -point mesh.

3. Choose the pseudopotentials and exchange-correlation functional.
4. Perform the self-consistent ground-state calculation and check the density, Kohn–Sham orbitals, eigenvalues, and band gap.
5. Store the ground-state data in a form that can be read as the initial state for the real-time calculation.

Octopus performs solid-state calculations using a real-space grid together with periodic boundary conditions. Since input syntax, output file names, and the locations of time-series data may depend on the version and compilation options, it is important to compare the input examples used here with the documentation of one’s own computational environment.

## 5.2 Time-Dependent Kohn–Sham Equation for Bloch Orbitals

In this subsection, we introduce the time-dependent Kohn–Sham equation solved when a spatially uniform time-dependent electric field is applied to a solid. The key point is that, in a periodic system, the length-gauge potential  $-e\mathbf{E}(t) \cdot \mathbf{r}$  breaks lattice periodicity. Therefore, it is convenient to use the velocity gauge, which is compatible with periodic boundary conditions.

When the wavelength of light is much longer than the size of the unit cell, the electric field felt by electrons in the solid can be approximated as spatially uniform:

$$\mathbf{E}(\mathbf{r}, t) \simeq \mathbf{E}(t). \quad (69)$$

This is called the long-wavelength approximation or the dipole approximation. For visible and near-infrared light, the wavelength is typically hundreds of nanometers or longer, whereas the size of a unit cell is typically a few Å. Thus, this approximation is widely used in first-principles electron dynamics calculations.

For an isolated system, a uniform electric field can be represented in the length gauge as

$$v_{\text{ext}}(\mathbf{r}, t) = -e\mathbf{E}(t) \cdot \mathbf{r}. \quad (70)$$

In a solid, however,  $\mathbf{r}$  is not a lattice-periodic quantity. If this term is directly inserted into the Kohn–Sham Hamiltonian, the periodic boundary condition is lost. Therefore, in real-time simulations of periodic systems, a uniform electric field is often represented by a vector potential in the velocity gauge.

In the velocity gauge, a spatially uniform electric field is represented by a vector potential  $\mathbf{A}(t)$ . We use the convention

$$\mathbf{E}(t) = -\frac{1}{c} \frac{d\mathbf{A}(t)}{dt}. \quad (71)$$

If the external field is absent in the remote past, then

$$\frac{\mathbf{A}(t)}{c} = -\int_{-\infty}^t dt' \mathbf{E}(t') \quad (72)$$

holds. In implementations using atomic units, the treatment of  $c$  and the sign convention for the electron charge may vary. Therefore, the same convention must be used consistently in input files, analysis scripts, and theoretical formulas.

In the velocity gauge, the external field appears through the replacement

$$\mathbf{p} \longrightarrow \mathbf{p} + \frac{e}{c} \mathbf{A}(t). \quad (73)$$

Here,  $e > 0$  is the elementary charge, and the charge of an electron is  $-e$ . The sign convention is tied to the definition of  $\mathbf{A}(t)$ , and the convention used by the code should be checked when analyzing results.

If one formally uses the length gauge, the time-dependent Kohn–Sham equation is written as

$$i\hbar \frac{\partial}{\partial t} \psi_m(\mathbf{r}, t) = \left[ \frac{\mathbf{p}^2}{2m} + \hat{v}_{\text{ion}} + v_{\text{H}}[\rho](\mathbf{r}, t) + v_{\text{xc}}[\rho](\mathbf{r}, t) - e\mathbf{E}(t) \cdot \mathbf{r} \right] \psi_m(\mathbf{r}, t). \quad (74)$$

However, the last term is incompatible with periodic boundary conditions. By applying a gauge transformation, the external field is instead included in the kinetic-energy term through the vector potential. The velocity-gauge Kohn–Sham equation becomes

$$i\hbar \frac{\partial}{\partial t} \tilde{\psi}_m(\mathbf{r}, t) = \left[ \frac{1}{2m} \left( \mathbf{p} + \frac{e}{c} \mathbf{A}(t) \right)^2 + \hat{v}_{\text{ion}}^{\mathbf{A}} + v_{\text{H}}[\rho](\mathbf{r}, t) + v_{\text{xc}}[\rho](\mathbf{r}, t) \right] \tilde{\psi}_m(\mathbf{r}, t). \quad (75)$$

Here,  $\hat{v}_{\text{ion}}^{\mathbf{A}}$  denotes the ionic potential including the vector-potential dependence. If the ionic potential is purely local,  $\hat{v}_{\text{ion}}^{\mathbf{A}} = \hat{v}_{\text{ion}}$ . If a nonlocal pseudopotential is used, however, the phase factor associated with the gauge transformation must be included.

A nonlocal pseudopotential does not generally commute with the position operator. Therefore, in the velocity gauge, it is not always sufficient to replace only  $\mathbf{p}$  by  $\mathbf{p} + e\mathbf{A}/c$ . In calculations using nonlocal pseudopotentials, the gauge correction to the nonlocal pseudopotential must be included consistently in both the time propagation and the current operator. This point should be checked carefully when using different codes or when implementing a method oneself.

Because the velocity-gauge Kohn–Sham Hamiltonian retains lattice periodicity, the time-dependent orbitals can still be written in Bloch form:

$$\tilde{\psi}_{b\mathbf{k}}(\mathbf{r}, t) = \exp[i\mathbf{k} \cdot \mathbf{r}] u_{b\mathbf{k}}(\mathbf{r}, t). \quad (76)$$

Here,  $u_{b\mathbf{k}}(\mathbf{r}, t)$  is lattice-periodic at each time.

Substituting Eq. (76) into Eq. (75) gives the time-dependent Kohn–Sham equation for the periodic part:

$$i\hbar \frac{\partial}{\partial t} u_{b\mathbf{k}}(\mathbf{r}, t) = \hat{h}_{\text{KS},\mathbf{k}}(t) u_{b\mathbf{k}}(\mathbf{r}, t). \quad (77)$$

The Hamiltonian is

$$\hat{h}_{\text{KS},\mathbf{k}}(t) = \frac{1}{2m} \left( \mathbf{p} + \hbar\mathbf{k} + \frac{e}{c} \mathbf{A}(t) \right)^2 + \hat{v}_{\text{ion},\mathbf{k}}^{\mathbf{A}} + v_{\text{H}}[\rho](\mathbf{r}, t) + v_{\text{xc}}[\rho](\mathbf{r}, t). \quad (78)$$

When a nonlocal pseudopotential is present,  $\hat{v}_{\text{ion},\mathbf{k}}^{\mathbf{A}}$  may be expressed conceptually as

$$\hat{v}_{\text{ion},\mathbf{k}}^{\mathbf{A}} = e^{-i\mathbf{K}(t) \cdot \mathbf{r}} \hat{v}_{\text{ion}} e^{i\mathbf{K}(t) \cdot \mathbf{r}}, \quad \mathbf{K}(t) = \mathbf{k} + \frac{e}{\hbar c} \mathbf{A}(t). \quad (79)$$

For a local potential, these phase factors cancel. For a nonlocal pseudopotential, however, they make an explicit contribution.

Equation (77) is the central equation solved in real-time TDDFT calculations for solids. The initial condition is given by the ground-state orbitals  $u_{b\mathbf{k}}(\mathbf{r}, 0)$ , and the occupied bands at each  $\mathbf{k}$  point are propagated in time under a given external field  $\mathbf{A}(t)$ .

**Common misconception** Using the velocity gauge does not change the physics. The length and velocity gauges are exactly equivalent when treated without approximation. In numerical calculations, however, finite basis sets, finite grids, pseudopotential approximations, and approximate time propagators are used. If gauge factors and current operators are not treated consistently, different gauges may appear to give different numerical results.

### 5.3 Current Density and Optical Response Functions

In this subsection, we explain how optical conductivity and dielectric functions are extracted from the current density obtained in a real-time calculation. For optical responses of solids, the current

density is a more natural quantity than the dipole moment. Intuitively, one applies an electric field, computes how electrons flow in the crystal, and Fourier-analyzes the time-dependent current to obtain the frequency-dependent response.

Let  $\Omega$  be the volume of the unit cell. The macroscopic current density is evaluated as

$$\mathbf{J}(t) = -\frac{e}{\Omega N_k} \sum_{\mathbf{k}} \sum_b f_{b\mathbf{k}} \int_{\Omega} d\mathbf{r} u_{b\mathbf{k}}^*(\mathbf{r}, t) \hat{\mathbf{v}}_{\mathbf{k}}(t) u_{b\mathbf{k}}(\mathbf{r}, t). \quad (80)$$

Here,  $\hat{\mathbf{v}}_{\mathbf{k}}(t)$  is the velocity operator, defined by

$$\hat{\mathbf{v}}_{\mathbf{k}}(t) = \frac{1}{i\hbar} \left[ \mathbf{r}, \hat{h}_{\text{KS},\mathbf{k}}(t) \right]. \quad (81)$$

If only local potentials are considered, the velocity operator is approximately

$$\hat{\mathbf{v}}_{\mathbf{k}}(t) \simeq \frac{1}{m} \left( \mathbf{p} + \hbar \mathbf{k} + \frac{e}{c} \mathbf{A}(t) \right).$$

However, when nonlocal pseudopotentials are used, the contribution from  $[\mathbf{r}, \hat{v}_{\text{ion},\mathbf{k}}^{\mathbf{A}}]$  must also be included.

When the response to a weak external field is linear, the current density and electric field are related by

$$J_a(t) = \sum_b \int_{-\infty}^t dt' \sigma_{ab}(t-t') E_b(t'). \quad (82)$$

Here,  $a$  and  $b$  denote Cartesian components, and  $\sigma_{ab}(t)$  is the optical conductivity tensor in the time domain. Fourier transformation gives

$$J_a(\omega) = \sum_b \sigma_{ab}(\omega) E_b(\omega). \quad (83)$$

Once the optical conductivity is obtained, the dielectric function is calculated as

$$\epsilon_{ab}(\omega) = \delta_{ab} + \frac{4\pi i}{\omega} \sigma_{ab}(\omega). \quad (84)$$

This expression is written in Gaussian units. In SI units, the prefactor is different and includes  $\epsilon_0$ . Therefore, the unit convention of the code output and that of the analysis script must be made consistent.

For molecules, the polarizability  $\alpha_{ab}(\omega)$  was evaluated by Fourier-transforming the dipole moment  $d_a(t)$ . For solids, one instead Fourier-transforms the unit-cell-averaged current  $J_a(t)$  to obtain the optical conductivity  $\sigma_{ab}(\omega)$ . This is one of the most important practical differences between molecular and solid-state real-time calculations.

In real-time linear-response calculations, it is convenient to apply a sufficiently weak impulsive electric field and analyze the subsequent free current oscillation. For example, an impulsive electric field in the  $b$  direction is written as

$$E_b(t) = E_0 \delta(t). \quad (85)$$

Using Eq. (71), the corresponding vector potential is a step function,

$$A_b(t) = -c E_0 \Theta(t). \quad (86)$$

Substituting Eq. (85) into Eq. (82), one obtains

$$J_a(t) = E_0 \sigma_{ab}(t). \quad (87)$$



Therefore, the optical conductivity can be obtained by Fourier-transforming the induced current:

$$\sigma_{ab}(\omega) = \frac{1}{E_0} \int_0^{T_{\text{prop}}} dt J_a(t) e^{i\omega t} e^{-\gamma t}. \quad (88)$$

Here,  $T_{\text{prop}}$  is the total propagation time, and  $\gamma$  is a damping width introduced to suppress oscillations due to the finite time cutoff. In practical analysis, one subtracts any stationary current or numerical offset before the Fourier transform and, if necessary, applies a window function.

Combining Eq. (88) with Eq. (84), one obtains the dielectric function  $\epsilon_{ab}(\omega)$  from a real-time TDDFT calculation. A useful feature of this approach is that a single time propagation can provide the linear response over a broad frequency range.

**Key point** A real-time linear-response calculation consists of three basic steps: apply a weak perturbation, compute the current, and Fourier-transform the current. If the perturbation is sufficiently weak, the resulting  $\sigma(\omega)$  and  $\epsilon(\omega)$  do not depend on the perturbation strength. If they change when the field strength is varied, nonlinear effects are already mixed into the response.

## 5.4 Linear-Response Calculations for Solids Using Octopus

In this subsection, we review the workflow for computing the linear optical response of a solid in real time using Octopus. We take bulk *Si* as an example. The Bloch orbitals obtained in the ground-state calculation are read as the initial state, a weak impulsive electric field is applied, and the dielectric function is extracted from the induced current.

Source code 8 shows an Octopus input file for computing the linear response of bulk *Si* under a weak impulsive electric field.

[https://shunsuke-sato.github.io/page/etc/lecture\\_notes/src\\_tddft\\_note/lin\\_res\\_si/inp\\_td](https://shunsuke-sato.github.io/page/etc/lecture_notes/src_tddft_note/lin_res_si/inp_td)

Source Code 8: Octopus input file: linear response of bulk *Si*

```

1
2 CalculationMode = td
3 FromScratch = yes
4 PeriodicDimensions = 3
5
6
7
8 %LatticeVectors
9 0.0 | 0.5 | 0.5
10 0.5 | 0.0 | 0.5
11 0.5 | 0.5 | 0.0
12 %
13
14 a = 10.18
15 %LatticeParameters
16 a | a | a
17 %
18
19 Spacing = (a/sqrt(2.0))/16.1
20
21 %ReducedCoordinates
22 "Si" | 0.0 | 0.0 | 0.0
23 "Si" | 1/4 | 1/4 | 1/4
24 %
25
26 nk = 5
27 %KPointsGrid
28 nk | nk | nk
29 %
30
31 KPointsUseSymmetries = yes
32 %SymmetryBreakDir
33 1 | 0 | 0
34 %
35
36
37 ExperimentalFeatures = yes

```

```

38 kick0 = 0.005
39 clight = 137.035999679 # in a.u.
40
41
42
43 %TDExternalFields
44 vector_potential | 1.0 | 0.0 | 0.0 | 0.0 | "kick_ac"
45 %
46
47 %TDFunctions
48 "kick_ac" | tdf_from_expr | "-kick0*clight*step(t)"
49 %
50
51 %TDOutput
52 total_current
53 %
54
55 TDPropagator = aetrs
56 TDExponentialMethod = taylor
57 TDExpOrder = 4
58 TDPropagationTime = 10*fs
59 TDTimeStep = 0.08

```

[https://shunsuke-sato.github.io/page/etc/lecture\\_notes/src\\_tddft\\_note/lin\\_res\\_si/run\\_td.sh](https://shunsuke-sato.github.io/page/etc/lecture_notes/src_tddft_note/lin_res_si/run_td.sh)

Source Code 9: Shell script for running the linear-response calculation of bulk  $Si$

```

1  #!/bin/bash
2
3  export OMP_NUM_THREADS=4
4  export MY_OCTOPUS="/work/sato/octopus_tutorial/octopus_execute/install/bin/octopus"
5
6  cp inp_td inp
7
8  mpirun -np 4 "${MY_OCTOPUS}" >log.td.log 2>&1

```

In this input file, the periodic Kohn–Sham orbitals obtained in the ground-state calculation are used as the initial state, and the system is propagated in time under a weak external field. The main points to check are as follows.

**CalculationMode** This specifies the real-time calculation mode. The calculation starts from the result of the ground-state calculation.

**TDPropagator** This specifies the approximation used for the time-evolution operator. As in molecular calculations, propagation schemes that respect time-reversal symmetry are often used.

**TDTimeStep** This specifies the time step. A sufficiently small time step is required to obtain reliable spectra at high energies.

**TDPropagationTime** This specifies the total propagation time. The frequency resolution is roughly determined by  $2\pi\hbar/T_{\text{prop}}$ .

**External field** In linear-response calculations, the impulsive field must be weak enough that the induced response is proportional to the field strength.

**$k$ -point sampling** The current response of a solid can be sensitive to the number of  $k$  points. This is especially important for low-frequency responses, metallic responses, and nonlinear current dynamics.

When the time-dependent calculation finishes successfully, the time evolution of the induced current is written to output files. The names of output files and the directory structure depend on the Octopus version and input settings, but time-series data for the current density and external field are typically stored in the output directory for the time-dependent calculation.

Figure 3 shows the time evolution of the current after an impulsive electric field is applied to bulk  $Si$ .

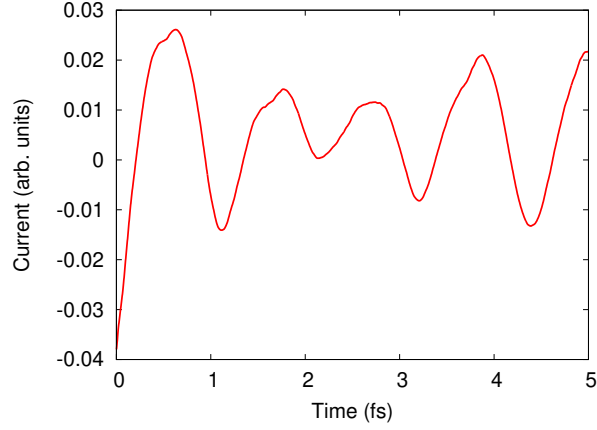


Figure 3: Time evolution of the current after an impulsive electric field is applied to bulk  $Si$ .

The oscillation in Fig. 3 reflects the time evolution of a superposition of electron-hole excitations created by the external field. Fourier analysis of this oscillating current gives the optical response of bulk  $Si$ .

Source code 10 shows a Python script for calculating the dielectric function from the time-dependent current.

[https://shunsuke-sato.github.io/page/etc/lecture\\_notes/src\\_tddft\\_note/lin\\_res\\_si/optical\\_conductivity\\_solids.py](https://shunsuke-sato.github.io/page/etc/lecture_notes/src_tddft_note/lin_res_si/optical_conductivity_solids.py)

Source Code 10: Python script for calculating the dielectric function of  $Si$

```

1  # /// script
2  # dependencies = [
3  #     "pandas",
4  #     "matplotlib",
5  #     "numpy"
6  # ]
7  # ///
8
9  import numpy as np
10 import pandas as pd
11 import matplotlib.pyplot as plt
12
13
14 def main():
15     filename = "td.general/total_current"
16
17     df = pd.read_csv(
18         filename,
19         comment="#",
20         sep=r"\s+",
21         header=None,
22         usecols=[0, 1, 2, 3, 4],
23         names=["Iter", "t", "jx", "jy", "jz"],
24         engine="python",
25     )
26
27     print("Data preview:")
28     print(df.head())
29     print("\nData shape:", df.shape)
30
31     cell_volume = 263.7445
32
33     tt = df["t"].to_numpy()
34     jxt = df["jx"].to_numpy() / cell_volume
35     jyt = df["jy"].to_numpy() / cell_volume
36     jzt = df["jz"].to_numpy() / cell_volume
37
38     if len(tt) < 2:
39         raise ValueError("Need at least two time points to compute dt.")
40
41     dt = tt[1] - tt[0]

```

```

42 kick_k = 0.005
43 kick_dir = np.array([1.0, 0.0, 0.0])
44
45
46 ev = 27.2114
47 nomega = 400
48 omega_ini = 0.1 / ev
49 omega_fin = 20.0 / ev
50 gamma = 0.3 / ev
51
52 omega = np.linspace(omega_ini, omega_fin, nomega)
53
54 sigma = np.zeros(nomega, dtype=complex)
55
56 for iw in range(nomega):
57     jxw = 0.0 + 0.0j
58     jyw = 0.0 + 0.0j
59     jzw = 0.0 + 0.0j
60
61     for it in range(len(tt)):
62         phase = np.exp((1j * omega[iw] - gamma) * tt[it])
63         jxw += phase * jxt[it]
64         jyw += phase * jyt[it]
65         jzw += phase * jzt[it]
66
67     sigma[iw] = -dt * (
68         kick_dir[0] * jxw
69         + kick_dir[1] * jyw
70         + kick_dir[2] * jzw
71     ) / kick_k
72
73 epsilon_w = 1.0 + 4.0j * np.pi * sigma / omega
74
75 print("\nFirst 10 omega values:")
76 print(omega[:10])
77
78 print("\nFirst 10 sigma values:")
79 print(sigma[:10])
80
81 plt.figure()
82 plt.plot(omega * ev, epsilon_w.real, label="Re(epsilon)")
83 plt.plot(omega * ev, epsilon_w.imag, label="Im(epsilon)")
84 plt.xlim(0.0, 20.0)
85 plt.ylim(-20, 20.0)
86 plt.xlabel("Omega (eV)")
87 plt.ylabel("epsilon")
88 plt.legend()
89 plt.tight_layout()
90 plt.savefig("dielectric_function.png")
91
92
93 if __name__ == "__main__":
94     main()

```

In this analysis, the Fourier transform corresponding to Eq. (88) is performed, and the dielectric function is then evaluated using Eq. (84). The resulting dielectric function is shown in Fig. 4.

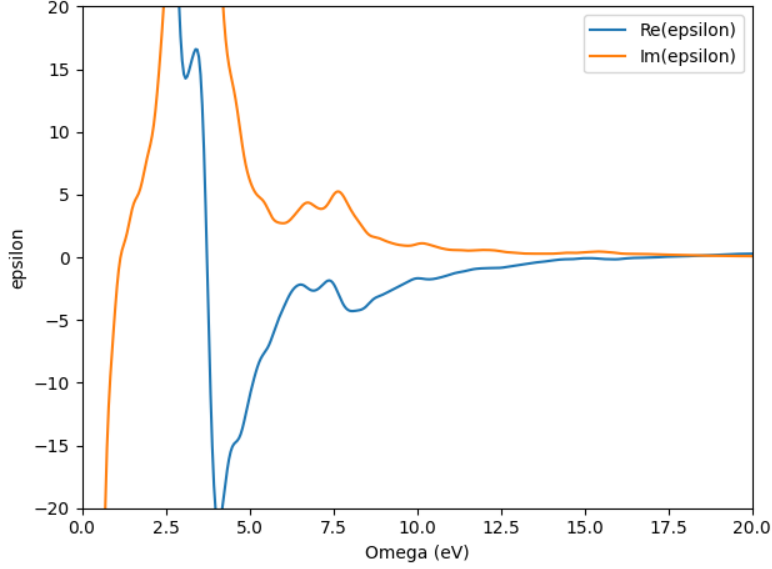


Figure 4: Dielectric function of bulk *Si* calculated by real-time TDDFT.

The peak structures in Fig. 4 correspond to electronic excitations from valence bands to conduction bands. However, the peak positions and intensities depend on the exchange-correlation functional, pseudopotentials, lattice constant, grid spacing, number of  $\mathbf{k}$  points, and damping width  $\gamma$ . Therefore, when making quantitative comparisons with experimental spectra, convergence with respect to these computational conditions must be checked.

## 5.5 Numerical Pump-Probe Calculations and Transient Optical Response

The real-time formulation can also be used to simulate pump-probe experiments. In such calculations, a strong pump pulse first drives the solid into a nonequilibrium electronic state, and a much weaker probe pulse is then used to measure the optical response of that transient state. This procedure is often called a numerical pump-probe experiment.

Let  $\mathbf{E}_{\text{pump}}(t)$  be the pump electric field and  $\mathbf{E}_{\text{probe}}(t - T_{\text{delay}})$  the probe electric field delayed by  $T_{\text{delay}}$ . One performs two real-time simulations under identical numerical conditions:

1. a pump-only calculation driven by  $\mathbf{E}_{\text{pump}}(t)$ ;
2. a pump-probe calculation driven by  $\mathbf{E}_{\text{pump}}(t) + \mathbf{E}_{\text{probe}}(t - T_{\text{delay}})$ .

The corresponding currents are denoted by  $\mathbf{J}_{\text{pump}}(t)$  and  $\mathbf{J}_{\text{pump+probe}}(t; T_{\text{delay}})$ . The probe-induced current is then defined as the difference

$$\mathbf{J}_{\text{probe}}(t; T_{\text{delay}}) = \mathbf{J}_{\text{pump+probe}}(t; T_{\text{delay}}) - \mathbf{J}_{\text{pump}}(t). \quad (89)$$

This subtraction isolates the current induced by the weak probe field in the presence of the nonequilibrium state created by the pump pulse.

If the probe field is sufficiently weak, the transient optical conductivity can be extracted as

$$\sigma_{ab}^T(\omega; T_{\text{delay}}) = \frac{\int dt J_{\text{probe},a}(t; T_{\text{delay}}) e^{i\omega t} e^{-\gamma t}}{\int dt E_{\text{probe},b}(t - T_{\text{delay}}) e^{i\omega t} e^{-\gamma t}}. \quad (90)$$

The corresponding transient dielectric function is

$$\epsilon_{ab}^T(\omega; T_{\text{delay}}) = \delta_{ab} + \frac{4\pi i}{\omega} \sigma_{ab}^T(\omega; T_{\text{delay}}). \quad (91)$$

The superscript “T” emphasizes that this quantity characterizes a transient, nonequilibrium state and may depend on the pump–probe delay time.

There are two important differences from ordinary linear-response calculations. First, after the pump pulse, the system is generally no longer in its ground state, and time-translation symmetry is lost. Therefore, the response can depend on both the observation time and the delay time. Second, for a strongly excited system, the extracted dielectric function can depend on the probe pulse shape and on the delay time, especially when coherent electronic motion remains. Thus, the transient dielectric function should be interpreted as an operational quantity defined by the simulated pump–probe protocol.

The probe field must be weak enough that  $\mathbf{J}_{\text{probe}}$  is linear in the probe amplitude. At the same time, the pump field may be strong and can induce nonlinear excitation, carrier generation, or high-field current dynamics. In practice, one checks the linearity with respect to the probe strength by repeating the calculation with different probe amplitudes while keeping the pump pulse fixed.

**Practical note** In a pump–probe calculation, the pump-only and pump–probe simulations must use exactly the same numerical parameters: grid spacing, time step, total propagation time,  $\mathbf{k}$ -point mesh, propagator, and output settings. Otherwise, the subtraction in Eq. (89) may contain numerical noise rather than only the probe-induced response.

## 5.6 High-Harmonic Generation

In this subsection, we discuss solid-state high-harmonic generation as a representative nonlinear optical response under a strong laser field. When a solid is irradiated by an intense periodic laser field, the current response contains not only the fundamental frequency  $\omega_0$  of the external field but also its integer multiples. This phenomenon is called high-harmonic generation. In real-time TDDFT, one propagates Eq. (77) nonperturbatively under a strong external field and evaluates the harmonic spectrum by Fourier-analyzing the resulting current  $\mathbf{J}(t)$ .

Source code 11 shows an Octopus input file for calculating high-harmonic generation in Si.

[https://shunsuke-sato.github.io/page/etc/lecture\\_notes/src\\_tddft\\_note/lin\\_res\\_si/inp\\_td\\_hhg](https://shunsuke-sato.github.io/page/etc/lecture_notes/src_tddft_note/lin_res_si/inp_td_hhg)

Source Code 11: Octopus input file: high-harmonic generation in Si

```

1 CalculationMode = td
2 FromScratch = yes
3 PeriodicDimensions = 3
4
5
6
7
8 %LatticeVectors
9   0.0 | 0.5 | 0.5
10  0.5 | 0.0 | 0.5
11  0.5 | 0.5 | 0.0
12 %
13
14 a = 10.18
15 %LatticeParameters
16   a | a | a
17 %
18
19 Spacing = (a/sqrt(2.0))/16.1
20
21 %ReducedCoordinates
22 "Si" | 0.0 | 0.0 | 0.0
23 "Si" | 1/4 | 1/4 | 1/4
24 %
25
26 nk = 5
27 %KPointsGrid
28   nk | nk | nk
29 %
30

```

```

31 KPointsUseSymmetries = yes
32 %SymmetryBreakDir
33 1 | 0 | 0
34 %
35
36
37 ExperimentalFeatures = yes
38
39 kick0 = 0.005
40 clight = 137.035999679 # in a.u.
41
42 Iau = sqrt(1e12/3.509470e16) #Conservion in a.u.
43 omega = 1.55/27.2114 # 3000nm -> 0.4133 eV -> 0.01518 Ha
44 omega_zero = 0.0/27.2114 # 3000nm -> 0.4133 eV -> 0.01518 Ha
45 clight = 137.035999679 # in a.u.
46 Tpulse = 10.0*fs
47
48 %TDEExternalFields
49 vector_potential | 1 | 0 | 0 | 0.0 | "pump"
50 %
51
52 %TDFunctions
53 "pump" | tdf_from_expr | "-Iau/omega*clight*sin(pi*t/Tpulse)^2*(1-step(t-Tpulse))*sin(omega*(
54 t-Tpulse/2))"
55 %
56
57 %TDOutput
58 total_current
59 %
60
61 TDPropagator = aetrs
62 TDEponentialMethod = taylor
63 TDExpOrder = 4
64 TDPropagationTime = 10*fs
65 TDTimeStep = 0.08

```

[https://shunsuke-sato.github.io/page/etc/lecture\\_notes/src\\_tddft\\_note/lin\\_res\\_si/run\\_td\\_hhg.sh](https://shunsuke-sato.github.io/page/etc/lecture_notes/src_tddft_note/lin_res_si/run_td_hhg.sh)

Source Code 12: Shell script for calculating high-harmonic generation in bulk  $Si$

```

1 #!/bin/bash
2
3 export OMP_NUM_THREADS=4
4 export MY_OCTOPUS="/work/sato/octopus_tutorial/octopus_execute/install/bin/octopus"
5
6 cp inp_td_hhg inp
7
8 mpirun -np 4 "${MY_OCTOPUS}" >log_td_hhg.log 2>&1

```

Figure 5 shows the time evolution of the current induced by a strong laser field.

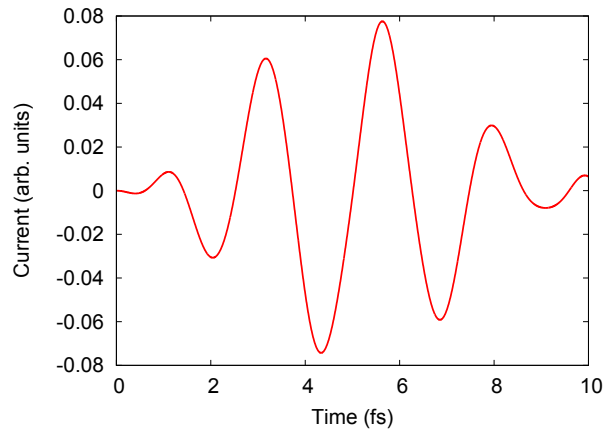


Figure 5: Time evolution of the current in a high-harmonic generation calculation.

The harmonic spectrum is evaluated by Fourier-transforming the current or its time derivative. For example, the emitted spectral intensity is conceptually related to the current spectrum as

$$I(\omega) \propto |\omega J(\omega)|^2. \quad (92)$$

The proportionality factor and the choice of window function depend on the physical quantity being analyzed and on the unit convention.

Source code 13 shows a Python script for calculating the high-harmonic spectrum. [https://shunsuke-sato.github.io/page/etc/lecture\\_notes/src\\_tddft\\_note/lin\\_res\\_si/spec\\_hhg.py](https://shunsuke-sato.github.io/page/etc/lecture_notes/src_tddft_note/lin_res_si/spec_hhg.py)

Source Code 13: Python script for calculating the high-harmonic spectrum of  $Si$

```

1  # /// script
2  # dependencies = [
3  #     "pandas",
4  #     "matplotlib",
5  #     "numpy"
6  # ]
7  # ///
8
9  import numpy as np
10 import pandas as pd
11 import matplotlib.pyplot as plt
12
13
14 def main():
15     filename = "td.general/total_current"
16
17     df = pd.read_csv(
18         filename,
19         comment="#",
20         sep=r"\s+",
21         header=None,
22         usecols=[0, 1, 2, 3, 4],
23         names=["Iter", "t", "jx", "jy", "jz"],
24         engine="python",
25     )
26
27     print("Data preview:")
28     print(df.head())
29     print("\nData shape:", df.shape)
30
31     cell_volume = 263.7445
32
33     tt = df["t"].to_numpy()
34     jxt = df["jx"].to_numpy() / cell_volume
35     jyt = df["jy"].to_numpy() / cell_volume
36     jzt = df["jz"].to_numpy() / cell_volume
37
38     if len(tt) < 2:
39         raise ValueError("Need at least two time points to compute dt.")
40
41     dt = tt[1] - tt[0]
42
43     kick_k = 0.005
44     kick_dir = np.array([1.0, 0.0, 0.0])
45
46     ev = 27.2114
47     nomega = 400
48     omega_ini = 0.1 / ev
49     omega_fin = 20.0 / ev
50     gamma = 0.3 / ev
51
52     omega = np.linspace(omega_ini, omega_fin, nomega)
53
54     spec = np.zeros(nomega)
55
56     for iw in range(nomega):
57         jxw = 0.0 + 0.0j
58         jyw = 0.0 + 0.0j
59         jzw = 0.0 + 0.0j
60
61         for it in range(len(tt)):
62             phase = np.exp((1j * omega[iw] - gamma) * tt[it])

```



```

63         jxw += phase * jxt[it]
64         jyw += phase * jyt[it]
65         jzw += phase * jzt[it]
66
67         spec[iw] = np.abs(dt*jxw)**2 + np.abs(dt*jyw)**2 + np.abs(dt*jzw)**2
68
69
70
71     plt.figure()
72     plt.plot(omega * ev, spec, label="HHG")
73     plt.xlim(0.0, 10.0)
74     plt.xlabel("Omega (eV)")
75     plt.ylabel("HHG intensity (arb. units)")
76     plt.yscale("log")
77     plt.legend()
78     plt.tight_layout()
79     plt.savefig("hbg_spec.png")
80
81
82 if __name__ == "__main__":
83     main()

```

The resulting harmonic spectrum is shown in Fig. 6.

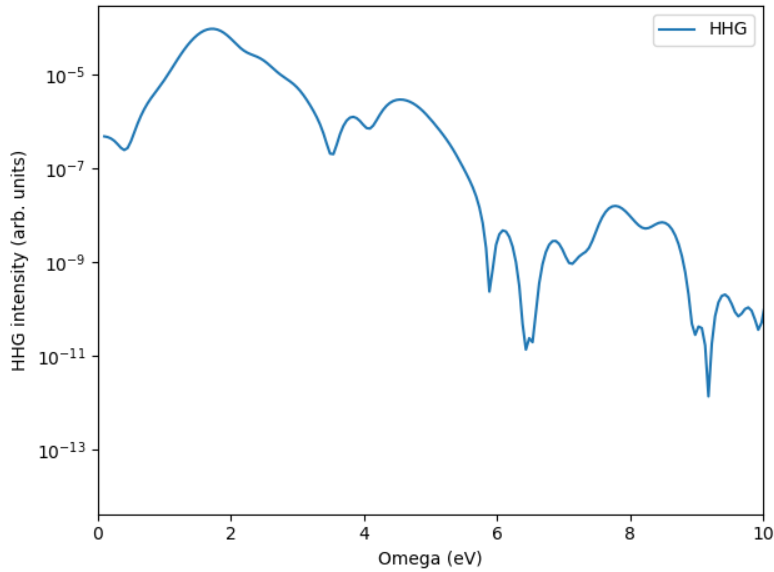


Figure 6: High-harmonic spectrum.

High-harmonic generation is a nonlinear response. Therefore, unlike in linear-response calculations, reducing the field strength does not leave the spectrum unchanged. The spectrum depends strongly on the field strength, pulse duration, polarization direction, number of  $\mathbf{k}$  points, time step, and description of the band gap. In strong fields, multiphoton absorption, tunneling excitation, intraband acceleration, and interband recombination may all contribute simultaneously, and the interpretation of the spectrum requires care.

**Physical interpretation** In solid-state high-harmonic generation, both intraband acceleration of electrons and interband dynamics of electron-hole pairs can be important. Therefore, the peaks in the spectrum cannot be interpreted simply by applying the three-step model of atomic gases without modification. Band structure, polarization direction, crystal symmetry, Berry curvature, and the number of excited electron-hole pairs can all influence the harmonic spectrum.

## 5.7 Choice of Numerical Parameters and Implementation Notes

In this subsection, we summarize numerical parameters that are especially important in real-time TDDFT calculations for solids. In solid-state calculations, errors can enter through the ground-state calculation, time propagation, Brillouin-zone integration, and Fourier analysis. Therefore, convergence should be checked not only for individual numerical parameters but also for the final physical quantity of interest.

Important numerical parameters include the following.

**Grid spacing** If the real-space grid is too coarse, Kohn–Sham orbitals and the spatial variation of pseudopotentials cannot be represented accurately. One should check convergence not only of the ground-state energy but also of the band gap, current response, and dielectric function.

**Number of  $k$  points** Optical responses of solids depend on Brillouin-zone integration. The  $k$ -point convergence is especially important for absorption edges of semiconductors, low-frequency responses of metals, and high-harmonic spectra. Strong-field currents and high harmonics may require denser  $k$  meshes than ordinary linear-response calculations.

**Time step** If the time step is too large, errors in the time-evolution operator increase, and high-frequency components of the current or energy conservation may become unreliable. Strong-field and high-harmonic calculations often require smaller time steps than weak-field linear-response calculations.

**Total propagation time** The frequency resolution of a linear-response spectrum is determined by the total propagation time. To resolve fine low-energy structures, long propagation times are necessary. In high-harmonic calculations, one must also decide how much of the free oscillation after the laser pulse should be included in the analysis.

**External-field strength** In linear-response calculations, the perturbation must be sufficiently weak. One should confirm that  $\sigma(\omega)$  and  $\epsilon(\omega)$  do not change when the field strength is varied. In nonlinear-response and high-harmonic calculations, the field strength itself is a physical control parameter, so its relation to experimental conditions must be considered carefully.

**Damping width and window function** Fourier transforms of finite-time signals contain oscillations due to truncation. Exponential damping and window functions smooth the spectrum, but they can also artificially modify peak widths and peak positions. One should check that the main conclusions do not depend on the chosen damping width.

**Exchange-correlation functional** For optical responses of semiconductors and insulators, the description of the band gap is important. Local-density and generalized-gradient approximations often underestimate band gaps, affecting the absorption edge. For quantitative comparison, the influence of the exchange-correlation functional and pseudopotentials should be assessed.

**Relaxation processes** Standard real-time calculations using adiabatic TDDFT often do not explicitly include irreversible relaxation due to electron–electron scattering or electron–phonon scattering. This limitation should be kept in mind when interpreting pump–probe responses at long delay times.

### Computational cautions

- Do not judge convergence from the ground-state total energy alone. For optical-response calculations, convergence of the band structure, current, dielectric function, and high-harmonic spectrum should be checked.

- If the impulsive field is too strong, the calculation no longer represents linear response. Confirm that the same  $\epsilon(\omega)$  is obtained when the field strength is changed.
- Do not mix Fourier-transform conventions, unit systems, or damping factors. Particular care is needed when converting among eV, Hartree, fs, and atomic units.
- In solid-state calculations using nonlocal pseudopotentials, the velocity operator contains a contribution from the nonlocal term. The current should not be evaluated as a simple expectation value of the canonical momentum alone.
- In pump–probe calculations, the pump-only and pump–probe calculations must be performed under identical numerical conditions. Otherwise, numerical errors may contaminate the difference current.

In practical applications, convergence should not be judged from a single physical quantity. For example, even if the total energy is converged, the high-energy part of the dielectric function or the high-harmonic spectrum may still be sensitive to the grid spacing or time step. Similarly, if the ground-state band structure is not well converged, the interpretation of peak positions and low-frequency responses obtained from real-time simulations becomes unreliable.

Finally, a real-time TDDFT calculation for solids may be viewed as a first-principles constitutive relation: the input is the external field  $\mathbf{E}(t)$  or  $\mathbf{A}(t)$ , and the output is the current  $\mathbf{J}(t)$ . From this viewpoint, linear response, nonlinear response, pump–probe response, and high-harmonic generation are all understood within the same time-dependent Kohn–Sham framework.

## References

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